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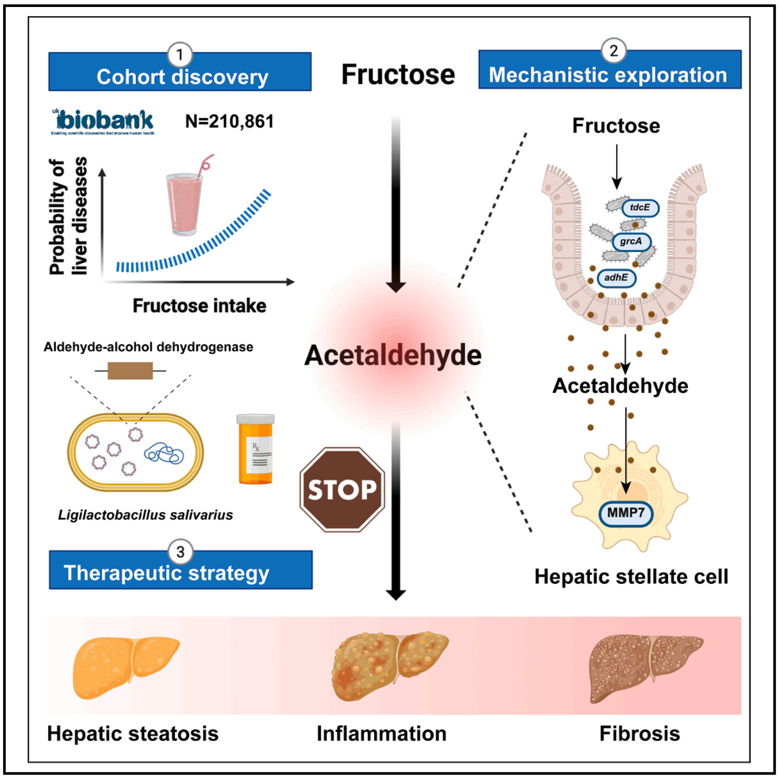
Targeting microbiota-generated acetaldehyde to prevent progression of metabolic dysfunction-associated steatotic liver disease

Graphical abstract



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In brief

Tang et al. reveal that high sugar intake promotes MASLD progression via microbiota-produced acetaldehyde, which drives fibrosis through MMP7 upregulation. An engineered probiotic, *Ligilactobacillus salivarius* HAM, enhances acetaldehyde clearance and prevents liver fibrosis in preclinical models, offering a novel therapeutic approach.

Highlights

- Excessive dietary sugar intake accelerates the progression from MASLD to MASH
- Gut microbiota in MASH patients produce elevated endogenous acetaldehyde
- Acetaldehyde promotes liver fibrosis by inducing MMP7 expression in stellate cells
- Engineered probiotics targeting acetaldehyde clearance prevent fibrosis effectively

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Targeting microbiota-generated acetaldehyde to prevent progression of metabolic dysfunction-associated steatotic liver disease

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SUMMARY

The progression from metabolic dysfunction-associated steatotic liver disease (MASLD) to steatohepatitis (MASH) entails rapid, often irreversible hepatic injury, underscoring the urgent need for innovative therapeutic strategies. Here, we demonstrate that excessive dietary sugar intake, particularly fructose, exacerbates liver disease progression through microbiota-mediated amplification of endogenous acetaldehyde production. Analysis of over 210,000 participants from the UK Biobank revealed a dose-dependent correlation between sugar consumption and liver-related mortality, accompanied by a microbial shift favoring acetaldehyde/ethanol fermentation pathways in MASH patients. Mechanistically, gut-derived acetaldehyde activates hepatic stellate cells via upregulation of matrix metalloproteinase-7 (MMP7), driving fibrogenesis. To mitigate this, we engineered *Ligilactobacillus salivarius* HAM, a probiotic strain with enhanced acetaldehyde-degrading capacity, which effectively halted fibrosis progression in preclinical models of diet-induced liver disease. These findings highlight microbiota-targeted modulation of aldehyde metabolism as a promising therapeutic avenue to intercept the transition from MASLD to MASH.

INTRODUCTION

Metabolic dysfunction-associated steatohepatitis (MASH), the progressive inflammatory stage of metabolic dysfunction-associated steatotic liver disease (MASLD), is a leading cause of liver-related morbidity and mortality worldwide. It is closely linked to rapid hepatic deterioration and often necessitates liver transplantation. Currently, approximately 30.69% of the global population is affected by MASLD, with an estimated 16.02% developing MASH, and these numbers are projected to increase as the epidemic expands.[1] MASH can progress to severe and irreversible liver conditions, including cirrhosis and hepatocellular carcinoma (HCC),[2] representing a significant clinical challenge due to the lack of effective therapies to halt its progression.

The transition from MASLD to MASH exhibits considerable heterogeneity, influenced by a complex interplay of factors such as lifestyle, metabolic comorbidities, and gut microbiota composition.[3] Among lifestyle factors, dietary nutrients, particularly sugar intake, have been strongly associated with MASLD onset,[4,5] yet their specific role in accelerating the progression to MASH remains poorly understood. Dysglycemia is also recognized as a key internal driver of disease progression.[6] Emerging evidence highlights disruptions in gut microbiota homeostasis and subsequent shifts in microbial metabolites as critical contributors to MASLD development,[7–9] suggesting a microbiome-mediated pathogenic axis.

In contrast to alcoholic liver disease (ALD), where exogenous alcohol intake is a primary factor, the contribution of endogenous alcohols and aldehydes produced by gut microbiota to MASLD pathogenesis has not been thoroughly explored. Recent observational studies have identified elevated levels of endogenous alcohols in the circulation of MASH patients, implicating microbial metabolism in disease progression.[10] This underscores the need for a systematic investigation into microbiota-mediated endogenous alcohol and aldehyde metabolism and their molecular mechanisms in the MASLD to MASH transition.

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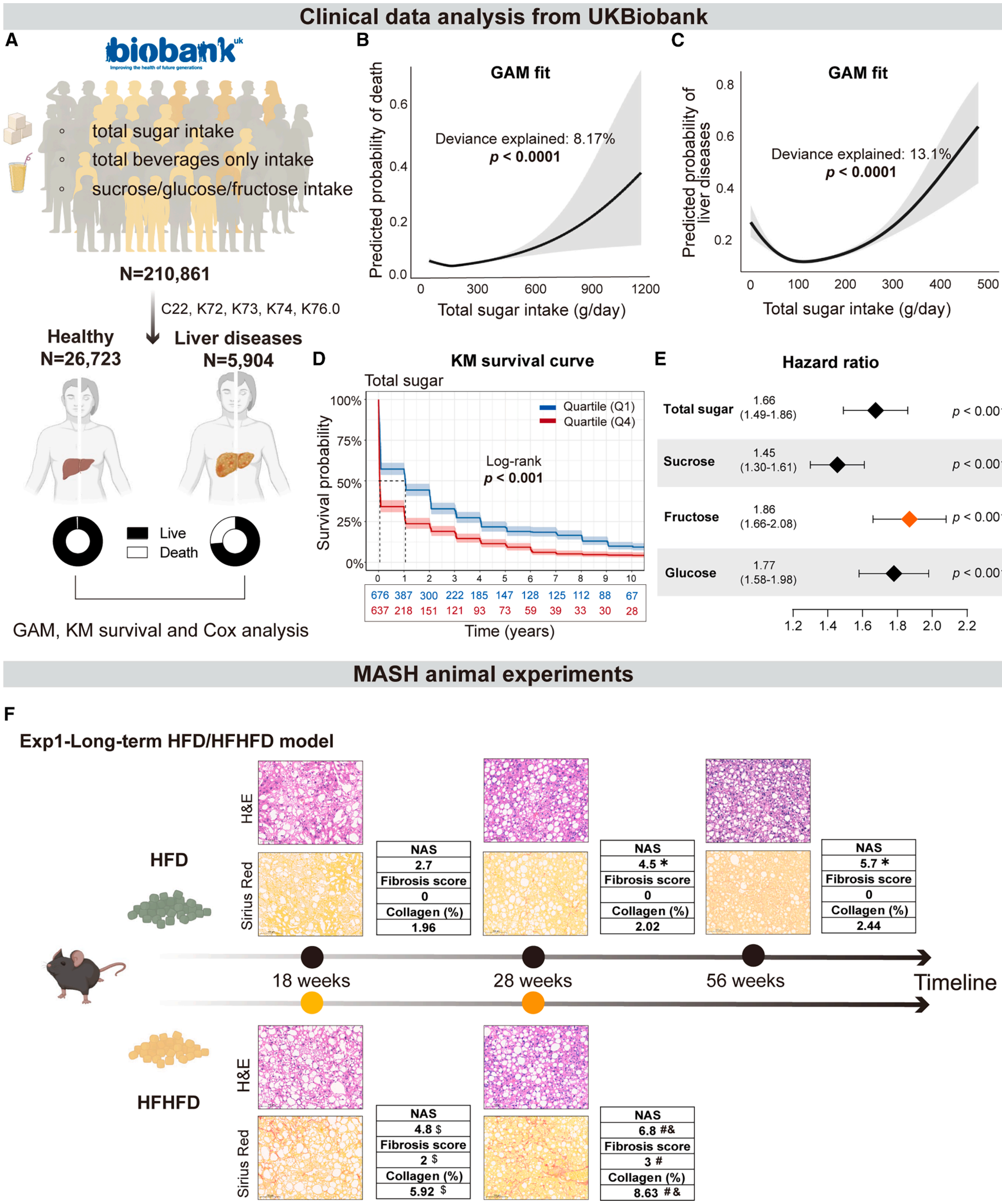

Figure 1. Sugar consumption was associated with the progression of MASLD-MASH

(A) Grouping of the UK Biobank cohort, including a healthy control group (n = 26,723) and a liver disease group (n = 5,904).

(B) GAM analysis, adjusted for age, sex, BMI, and smoking status, between total sugar intake and the possibility of death.

(C) GAM analysis, adjusted for age, sex, BMI, and smoking status, between total sugar intake and the incidence of liver diseases.

(D) Kaplan-Meier (KM) survival curves between the highest and lowest total sugar intake groups (Q1 and Q4) in the liver disease group.

(E) Hazard ratios for liver-related death were estimated using a univariate Cox proportional hazards model.

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In this context, we investigated the role of microbiota-driven endogenous acetaldehyde in the progression from MASLD to MASH. We demonstrate that excessive dietary fructose accelerates disease progression through microbiota-mediated production of endogenous acetaldehyde. The acetaldehyde activates hepatic stellate cells (HSCs) by upregulating matrix metalloproteinase-7 (MMP7), thereby promoting fibrosis in mouse and cell models. To counteract this pathogenic pathway, we engineered a probiotic strain, *Ligilactobacillus salivarius* (*L. salivarius*) HAM, originally isolated from fecal samples of healthy alcohol consumers, with enhanced capacity to degrade acetaldehyde through the overexpression of microbial bifunctional aldehyde-alcohol dehydrogenase. Therapeutic administration of this engineered probiotic effectively mitigated hepatic fibrosis and inflammation in preclinical models. Our findings identify microbiota-derived endogenous acetaldehyde as a key driver of MASLD progression and propose microbiota-targeted modulation of aldehyde metabolism as a promising therapeutic strategy for preventing and treating MASH.

RESULTS

Dietary sugar intake correlates with accelerated MASLD-MASH progression in humans and mice

To elucidate the impact of dietary sugar on liver disease progression, we analyzed data from 210,861 UK Biobank participants, focusing on nutrient intake and clinical outcomes (Figure 1A). Liver disease diagnoses (ICD-10 codes: C22, K72, K73, K74, and K76.0) identified 5,904 individuals with liver disease, contrasted with 26,723 healthy controls (Table S1). Using generalized additive models (GAMs), adjusted for age, sex, body mass index (BMI), and smoking status, we uncovered a nonlinear, J-shaped relationship between total sugar intake and mortality risk, with higher sugar consumption associated with increased mortality, paralleling with individual sugars (sucrose, fructose, and glucose) (Figures 1B and S1A–S1C). Similar dose-dependent associations emerged for the incidence of liver diseases, consistent across individual sugars (Figures 1C and S1D–S1F). Stratification into quartiles revealed that participants in the highest intake group (Q4, >154.9 g/day) had significantly reduced 10-year liver-related survival compared with the lowest intake group (Q1, <88.0 g/day) (Figures 1D and S1G–S1I). Cox proportional hazards analysis further revealed that dietary fructose was associated with a significantly increased risk of liver disease-related mortality compared with other individual sugars (Figure 1E).

Clinically, elevated sugar intake, particularly fructose, correlated with faster progression of liver pathology. To experimentally validate these findings, we employed mouse models fed high-fat diet (HFD) with or without fructose supplementation. Histological analyses (Sirius Red staining) demonstrated markedly accelerated fibrosis in mice on high-fat, high-fructose diet (HFHFD), with early fibrosis evident at 18 weeks and advanced bridging fibrosis by 28 weeks, surpassing HFD-fed counterparts (Figures 1F and S1J). Quantitative assessments confirmed greater collagen deposition and higher NAFLD activity score (NAS) in HFHFD mice, supporting a pathogenic role for dietary sugars in MASLD-MASH progression. By the 56th week, fibrosis progression in HFD-fed mice was modest compared with observations at 28 weeks (Figure 1F). To rigorously control for potential confounding effects of nutrient variability, we performed a validation experiment using a standardized diet comprising 60% kcal from fat, supplemented with 10% fructose in the drinking water. Consistent with our initial findings, histological analyses at 28 weeks demonstrated that fructose supplementation significantly exacerbated hepatic steatosis and accelerated fibrosis progression compared with HFD alone (Figure S1K).

Gut microbial remodeling toward acetaldehyde production drives MASLD-MASH progression

Given the gut microbiota's influence on liver pathology, we used antibiotic-treated mice to assess microbiota dependency. HFHFD-fed mice treated with broad-spectrum antibiotics exhibited minimal fibrosis, contrasting with severe collagen accumulation in untreated HFHFD mice, with bridging fibrosis (Figures 2A–2C). This indicated that microbiota are essential mediators of diet-induced MASLD-MASH progression.

To elucidate the gut microbial functional alterations underlying disease progression, we analyzed 146 fecal metagenomic datasets from the European NAFLD registry,[11] comprising biopsy-confirmed diagnoses of 20 MASLD, 50 MASH-0 (inflammatory stage), and 76 MASH-1 (fibrosis stage) patients (Figure 2D; Table S2A). Pathway profiling of gut microbiomes using the MetaCyc database revealed increased relative abundance through glycolytic pathways leading to pyruvate (PWY-5484, glycolysis II). Notably, downstream routes involved in acetaldehyde and ethanol biosynthesis (PWY-5480, pyruvate fermentation to ethanol I) were significantly enriched in MASH stages (Figure 2E). Random forest classification identified "pyruvate fermentation to ethanol I" as a top discriminative pathway distinguishing MASH from earlier stages (MASLD and MASH-0) (Figure 2F). Principal coordinate analysis (PCoA) based on weighted UniFrac distances demonstrated distinct microbial community structure clustering among MASLD, MASH-0, and MASH-1 groups (Figure 2G). Among the bacterial species with increased abundance in MASH (Table S2B), the majority harbored genes associated with pyruvate fermentation to ethanol I, further supporting a functional shift toward acetaldehyde and ethanol production (Figure 2H). These findings indicate that gut microbial metabolic capacity in MASH patients is significantly reprogrammed toward pathways generating acetaldehyde and ethanol. Importantly, a 28-week mouse model confirmed these microbial functional shifts, reinforcing their relevance to disease progression (Figures S2A–S2C).

(F) Representative H&E staining and Sirius Red staining of liver sections from animal experiment 1, long-term HFD/HFHFD model (6 biological replicates for each group); scale bars, 100 μm.

Data are presented as the mean ± SEM. *p* values for smooth terms were calculated using approximate Wald tests (B and C). *p* values for survival differences between groups were computed using the log-rank test (D). *p* values for hazard ratios were derived from Wald tests (E). *$p < 0.05$ compared with the 18-week HFD group based on the Kruskal-Wallis test with Benjamini-Hochberg adjustment, \$$p < 0.05$ compared with the 18-week HFD group based on the Mann-Whitney U test, #$p < 0.05$ compared with the 28-week HFD group based on the Mann-Whitney U test, and &$p < 0.05$ compared with the 18-week HFHFD group based on the Mann-Whitney U test (F). (A) was created with BioRender.com.

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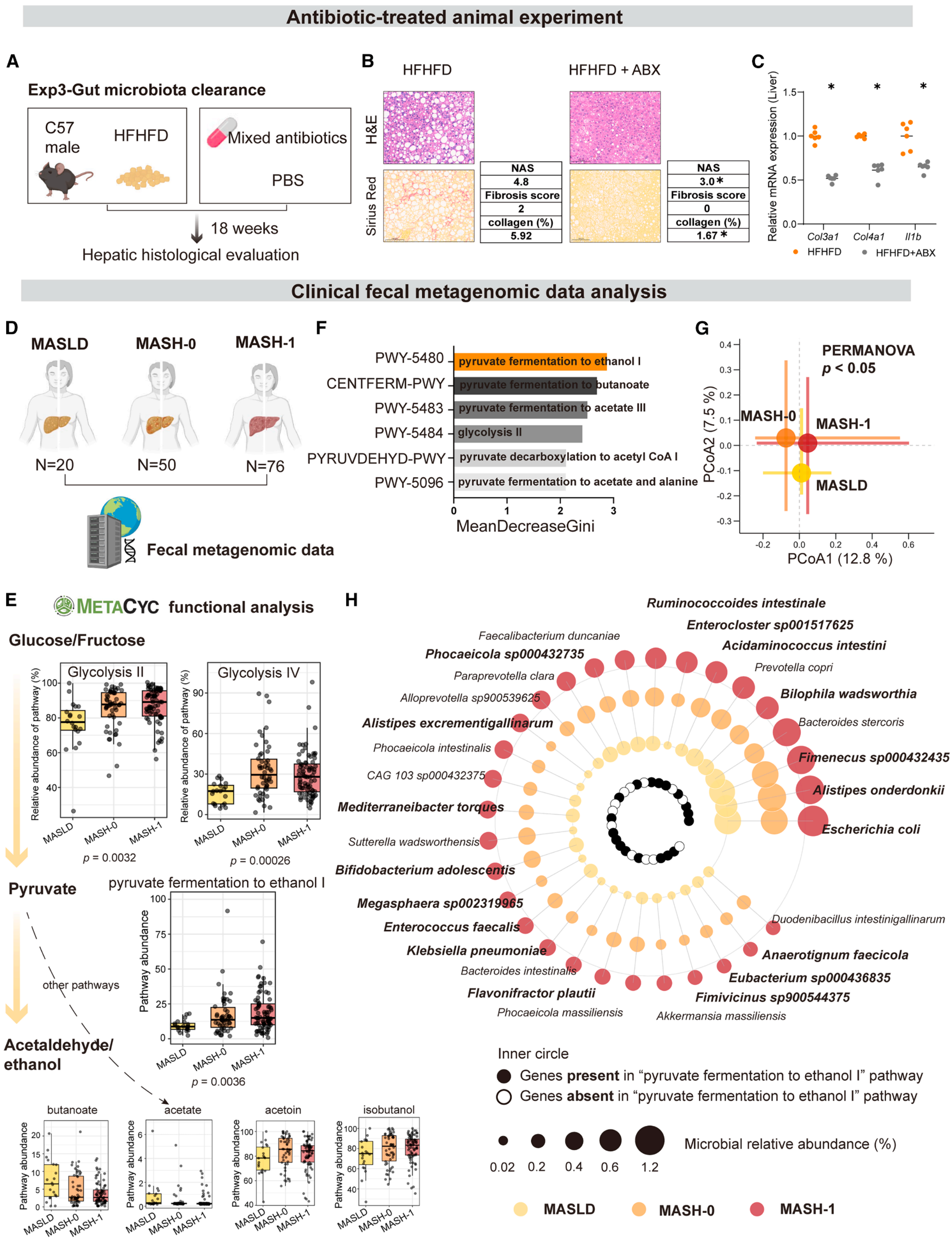

Figure 2. Gut microbial functional shift in MASLD-MASH progression

(A–C) Animal experiment 3, HFHFD + ABX experiment (6 biological replicates for each group) (A); representative images of H&E and Sirius Red staining of the liver and NAS based on the histological analysis (scale bars, 100 μm) (B); and mRNA expression of *Col3a1*, *Col4a1*, and *Il1b* of HFHFD and HFHFD + ABX groups (C).

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To validate the metagenomic findings, we assessed the *in vitro* capacity of fecal samples from clinical cohorts and mouse models to produce acetaldehyde from fructose and glucose (Figure 3A). The clinical cohort included 25 participants, comprising 5 healthy controls, 8 MASLD patients, and 12 MASH patients, stratified using established noninvasive scoring systems such as the NAFLD fibrosis score (NFS), fibrosis-4 (FIB-4), and hepatic steatosis index (HSI)[12–14] (Table S3). Fecal incubation with fructose or glucose revealed a marked increase in acetaldehyde in MASH samples—3.8-fold with fructose and 3.9-fold with glucose—compared with MASLD samples (Figure 3B). Similarly, feces from HFHFD-fed mice produced 2.4-fold more acetaldehyde with fructose and 1.6-fold with glucose than HFD-fed mice, indicating microbial functional shifts during disease progression (Figure 3C). Furthermore, longitudinal assessment of fecal acetaldehyde levels demonstrated early and sustained accumulation in HFHFD-fed mice. Notably, acetaldehyde concentrations were significantly elevated in HFHFD-fed mice compared with HFD as early as 3 weeks and persisted throughout the experimental timeline (Figures 3D and 3E). To further investigate the relationship between acetaldehyde and disease severity, we performed Spearman correlation analysis, which revealed a significant positive correlation between fecal acetaldehyde levels and NAS of mice across the 3-, 8-, and 28-week time points (Figure 3F). Validating these findings in our clinical cohort, we observed that patients with MASH exhibited significantly higher fecal acetaldehyde levels compared with MASLD (Figure 3G). Importantly, these fecal acetaldehyde concentrations also showed a positive correlation with the FIB-4 index (Figure 3H), indicating that gut-derived acetaldehyde accumulation is closely associated with the progression of hepatic fibrosis and disease severity in both murine and human patients. We also examined the ethanol produced during the fecal biotransformation process and observed significant changes in its concentration. For ethanol in MASH samples compared with MASLD samples, it was 3.0-fold with fructose and 2.4-fold with glucose (Figure S3A). Similarly, feces from HFHFD-fed mice produced 1.5-fold more acetaldehyde with fructose and 1.7-fold with glucose than HFD-fed mice (Figure S3B). Therefore, the data indicated that acetaldehyde and ethanol were significantly enriched and produced by the microbiota in both MASLD patients and HFHFD-induced animal models.

Dissecting the pathogenic origin of acetaldehyde from gut microbiota

Acetaldehyde accumulation stems from two distinct metabolic origins: host intestinal/hepatic metabolism and gut microbial fermentation. In the host, acetaldehyde is primarily generated during ethanol oxidation, a process catalyzed by alcohol dehydrogenase (ADH) and subsequently detoxified to acetate by aldehyde dehydrogenase 2 (ALDH2). By contrast, gut bacteria possess a distinct biosynthetic pathway capable of generating acetaldehyde from metabolic intermediates such as pyruvate (Figure 4A).

To delineate the specific contribution of this microbial pathway to disease progression, we performed metatranscriptomic profiling on fecal samples from our clinical cohort. Gene expression analysis revealed a significant transcriptional regulation of this pyruvate-to-acetaldehyde pathway in MASH (Figure 4B). Specifically, the transcriptional abundance (transcripts per million, TPM) of the key genes driving this pathway—*tdcE*, *grcA*, and *adhE*—was markedly elevated in the MASH patients compared with MASLD.

Consistent with the transcriptional activation of the microbial pyruvate-to-acetaldehyde pathway, we observed a substantial accumulation of acetaldehyde along the gut-liver axis. Specifically, acetaldehyde concentrations were significantly elevated in the small intestinal contents and tissues (distal small intestine and colon) of HFHFD-fed mice compared with HFD controls (Figures 4D and 4E). Crucially, this gut-derived metabolite was translocated into the circulation, as evidenced by markedly increased acetaldehyde levels in the portal vein blood (Figure 4C). Consequently, this led to significant acetaldehyde retention in hepatic tissues, but not in peripheral serum (Figure 4F).

To determine whether host metabolism contributed to this accumulation, we examined the expression of key enzymes involved in alcohol metabolism, *Adh1* (acetaldehyde production) and *Aldh2* (acetaldehyde clearance), in intestinal and hepatic tissues. The quantitative PCR (qPCR) analyses revealed no significant differences in the mRNA levels of these enzymes in the proximal/distal small intestine, colon (Figures 4G and 4H), or liver (Figure 4I) between the two groups.

To further rule out the possibility that acetaldehyde accumulation resulted from enhanced ethanol oxidation by the host, we quantified ethanol levels across multiple compartments. Strikingly, despite the surge in acetaldehyde, ethanol concentrations remained unchanged in intestinal contents, intestinal tissues, the portal vein, the liver, and serum (Figures S4A–S4E). This uncoupling of acetaldehyde from ethanol levels strongly suggests that the accumulation is largely oriented from microbial conversion, rather than the host ethanol oxidation pathway.

This conclusion was definitively substantiated by mechanistic interventions. Although inhibiting host ADH activity using 4-methylpyrazole (4-MP) significantly increased ethanol levels (Figure S4F), it failed to reduce hepatic acetaldehyde levels in HFHFD-fed mice (Figure 4J), confirming that the excess

(D) Grouping of European NAFLD registry cohort, including MASLD group (n = 20), MASH-0 group (inflammatory stage) (n = 50), and MASH-1 group (fibrosis stage) (n = 76).
(E) MetaCyc metabolic pathways analysis of the European NAFLD registry cohort gut microbiomes.
(F) Random forest modeling based on the relative abundance of MetaCyc metabolic pathways from the European NAFLD registry cohort gut microbiomes.
(G) PCoA analysis of the relative abundance of microbial species from the European NAFLD registry cohort gut microbiomes.
(H) The abundance of top differential microbial species among the MASLD, MASH-0, and MASH-1 groups, coupled with the presence or absence of the pyruvate fermentation to ethanol I pathway.
Data are presented as the mean ± SEM. *$p < 0.05$ compared with the HFHFD group based on the Mann-Whitney U test (B and C). p values were calculated by the Kruskal-Wallis test with Benjamini-Hochberg adjustment to assess differences across all groups (E). The p value was acquired by the PERMANOVA method (G). (A) and (D) were created with BioRender.com.

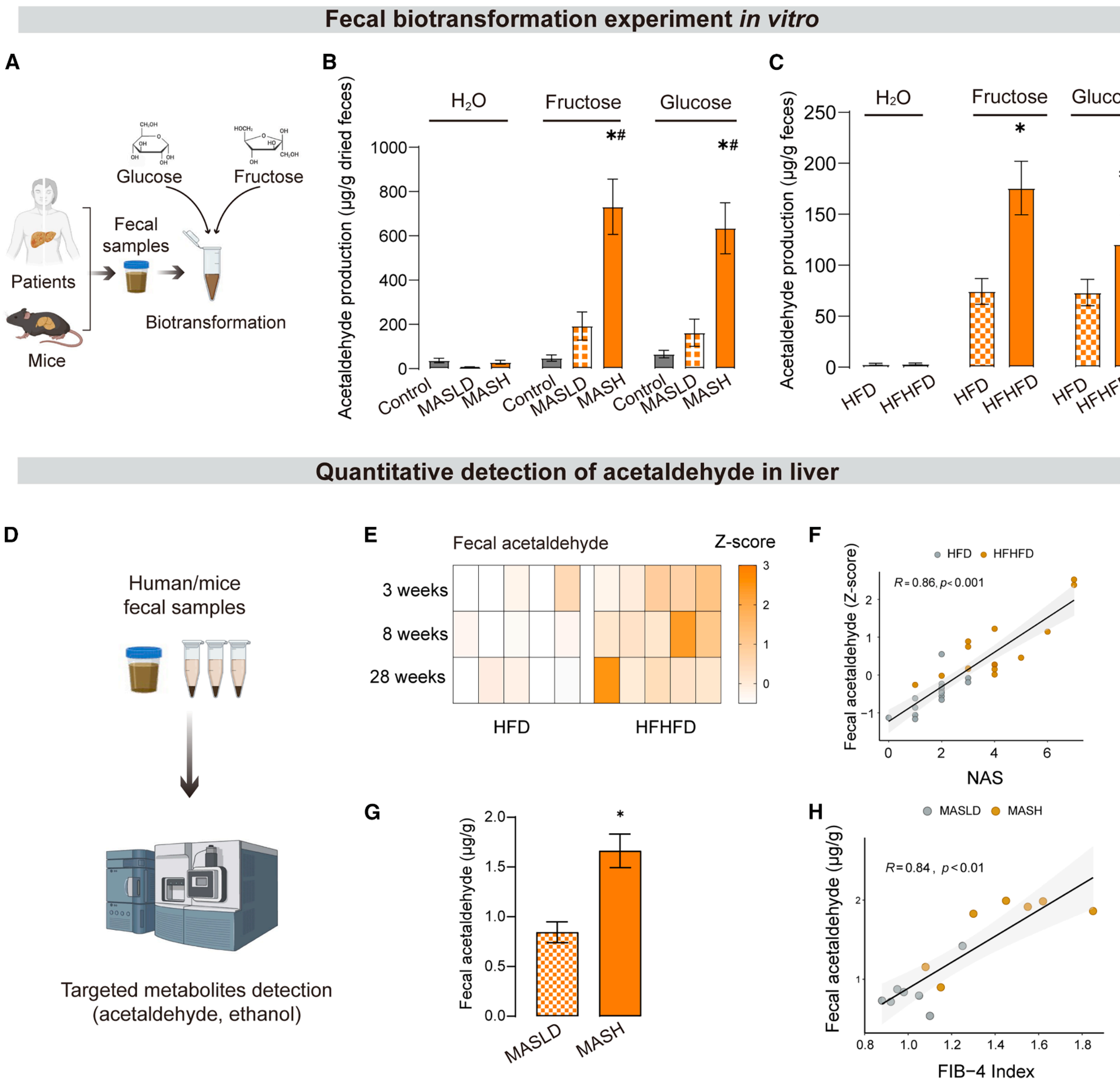

Figure 3. ***In vitro*** **transformation experiments of endogenous acetaldehyde by gut microbiota**

(A) *In vitro* fecal transformation experiments employing glucose or fructose as a carbon source and using feces from clinical cohorts and mouse experiments.

(B) Levels of acetaldehyde yielded in fecal transformation experiments employing feces from clinical cohorts (n = 5, 8, and 12 biological replicates for the control, MASLD, and MASH groups, respectively).

(C) Levels of acetaldehyde yielded in fecal transformation experiments employing feces of 18-week-old mice from animal experiment 1 (5 biological replicates for each group).

(D) Quantification of levels of acetaldehyde and ethanol using ultra-performance liquid chromatography with ultraviolet detection (UPLC-UV) and gas chromatography-mass spectrometry (GC-MS).

(E and F) Fecal acetaldehyde levels of HFD- and HFHFD-fed mice throughout modeling in animal experiment 1 (5 biological replicates for each group) (E) and Spearman correlation analysis between fecal acetaldehyde levels and NAS in animal experiment 1 (5 biological replicates for each group) (F).

(G and H) Fecal acetaldehyde levels of MASLD and MASH patients (7 biological replicates for each group) (G) and Spearman correlation analysis between fecal acetaldehyde levels and FIB-4 index in patients (7 biological replicates for each group) (H).

Data are presented as the mean ± SEM. $^{*,\#}p < 0.05$ compared with the healthy control group and MASLD patients, respectively, based on the Kruskal-Wallis test with Benjamini-Hochberg adjustment (B). $^{*}p < 0.05$ compared with the 18-week HFD group based on the Mann-Whitney U test (C). R value indicates the Spearman correlation coefficient, and $p < 0.001$ (F) and $p < 0.01$ (H) indicate the statistical significance based on the Spearman correlation. $^{*}p < 0.05$ compared with the MASLD patients based on the Mann-Whitney U test (G). (A) and (D) were created with BioRender.com.

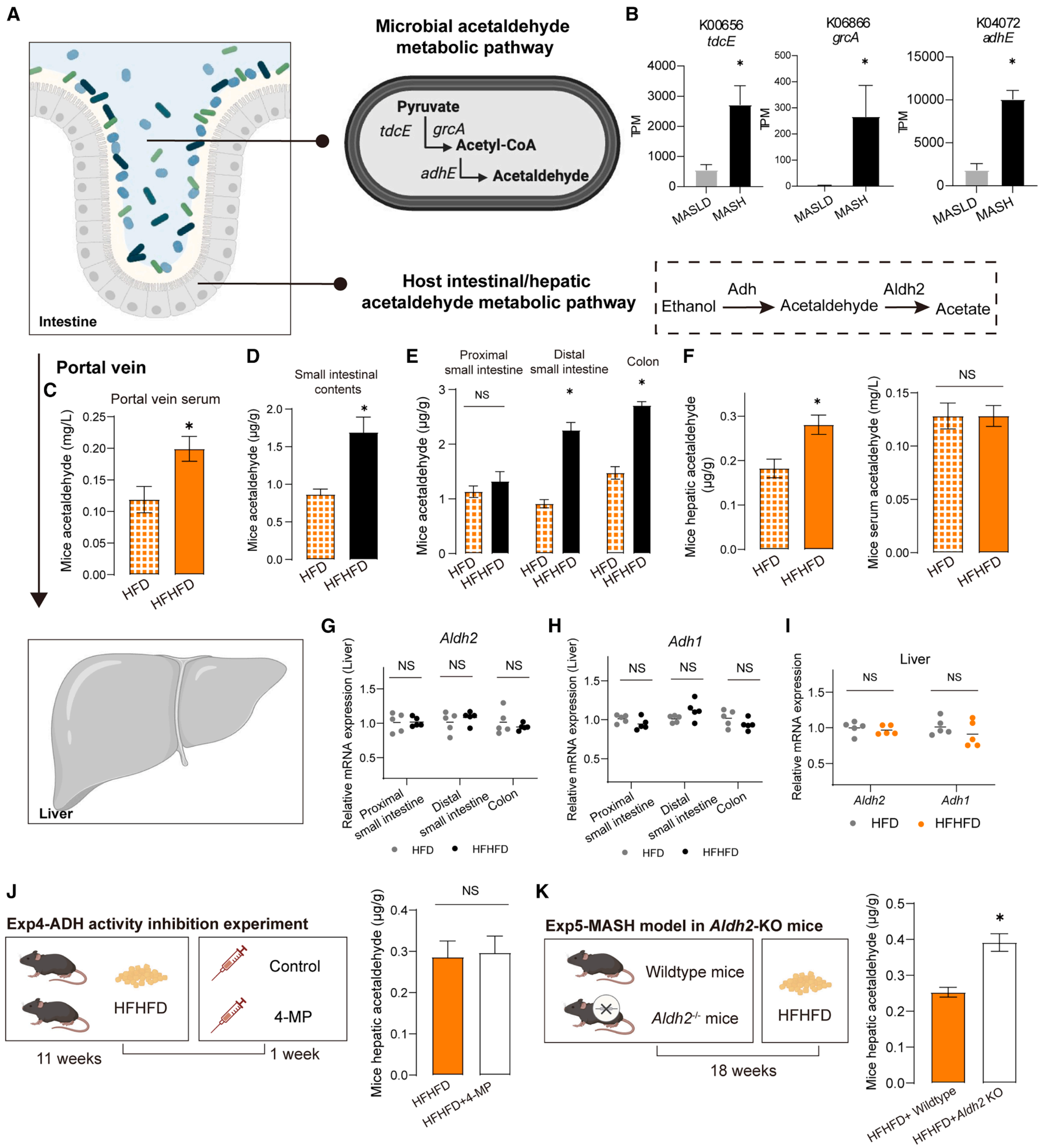

Figure 4. Endogenous acetaldehyde is derived from gut microbiota

(A) Acetaldehyde production pathway in gut microbiota and host.

(B) Gene expression analysis of *tdcE*, *grcA*, and *adhE* by metatranscriptomic profiling on fecal samples from the clinical cohort (n = 5 and 10 biological replicates for MASLD and MASH groups, respectively).

(C–F) Acetaldehyde levels of 18-week-old mice fed HFD or HFHFD in animal experiment 1 (5 biological replicates for each group), portal vein acetaldehyde levels (C), acetaldehyde levels of small intestinal contents of HFD- and HFHFD-fed mice (D), acetaldehyde levels of different intestinal parts of HFD- and HFHFD-fed mice (E), and acetaldehyde levels of the liver and serum of HFD- and HFHFD-fed mice (F).

(G and H) Intestinal mRNA expression of *Aldh2* (G) and *Adh1* (H) of 18-week-old mice fed either an HFD- or HFHFD in animal experiment 1 (5 biological replicates for each group).

(I) mRNA expression of hepatic *Aldh2* and *Adh1* of 18-week-old mice fed either an HFD or HFHFD in animal experiment 1 (5 biological replicates for each group).

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acetaldehyde was not generated via host ethanol oxidation. Conversely, disrupting the host's clearance capacity using whole-body *Aldh2* knockout mice resulted in a further, significant spike in hepatic acetaldehyde accumulation (Figure 4K), while ethanol concentrations remained unchanged (Figure S4G). Collectively, these findings provide compelling evidence that the excessive acetaldehyde in MASH is primarily of gut microbial origin, instead of host ethanol metabolism or systemic ethanol load.

Acetaldehyde promotes fibrosis through stellate cell activation and MMP7 upregulation

To elucidate the mechanistic role of endogenous acetaldehyde in driving hepatic inflammation and fibrosis, we administered low-dose acetaldehyde (50 mg/kg/day) orally via drinking water to mice fed an HFD for 18 weeks (Figure 5A). Histological assessment revealed markedly increased hepatic inflammation and collagen deposition in acetaldehyde-treated mice compared with HFD controls (Figure 5B). The qPCR results demonstrated significant upregulation of pro-inflammatory genes (*Tnfa* and *Il1b*) and pro-fibrotic genes (*Col1a1*, *Col3a1*, *Col4a1*, *Tgfb*, and *Timp1*) following acetaldehyde intervention (Figure 5C).

To assess the specific impact of impaired hepatic acetaldehyde clearance, we employed a liver-specific knockdown strategy using adeno-associated virus (AAV) vectors with thyroxine-binding globulin (TBG) promoter. Following 20 weeks of HFHFD feeding, AAV-sh*Aldh2* mice exhibited exacerbated hepatic fibrosis, evidenced by increased collagen accumulation and higher expression of pro-inflammatory and pro-fibrotic markers compared with AAV-shControl counterparts (Figures 5D–5F).

To identify molecular mediators, we performed RNA sequencing (RNA-seq) on liver tissues from two models: a 28-week cohort comparing control, HFD, and HFHFD groups and an 18-week cohort of HFD with or without acetaldehyde supplementation. Cross-comparison revealed 730 common differentially expressed genes (DEGs) across these conditions (Figure 5G). Principal component analysis and volcano plots confirmed significant transcriptomic shifts (Figures S5A–S5D). Gene Ontology (GO) enrichment implicated pathways related to collagen metabolism and extracellular matrix (ECM) organization (Figure 5H). Kyoto Encyclopedia of Genes and Genomes (KEGG) pathway analysis via gene set enrichment analysis (GSEA) highlighted enrichment in ECM-receptor interactions (specific interactions between ECM and cell-surface-associated components) in the HFHFD group and the HFD + acetaldehyde group relative to their respective controls (Figures S5E–S5G).

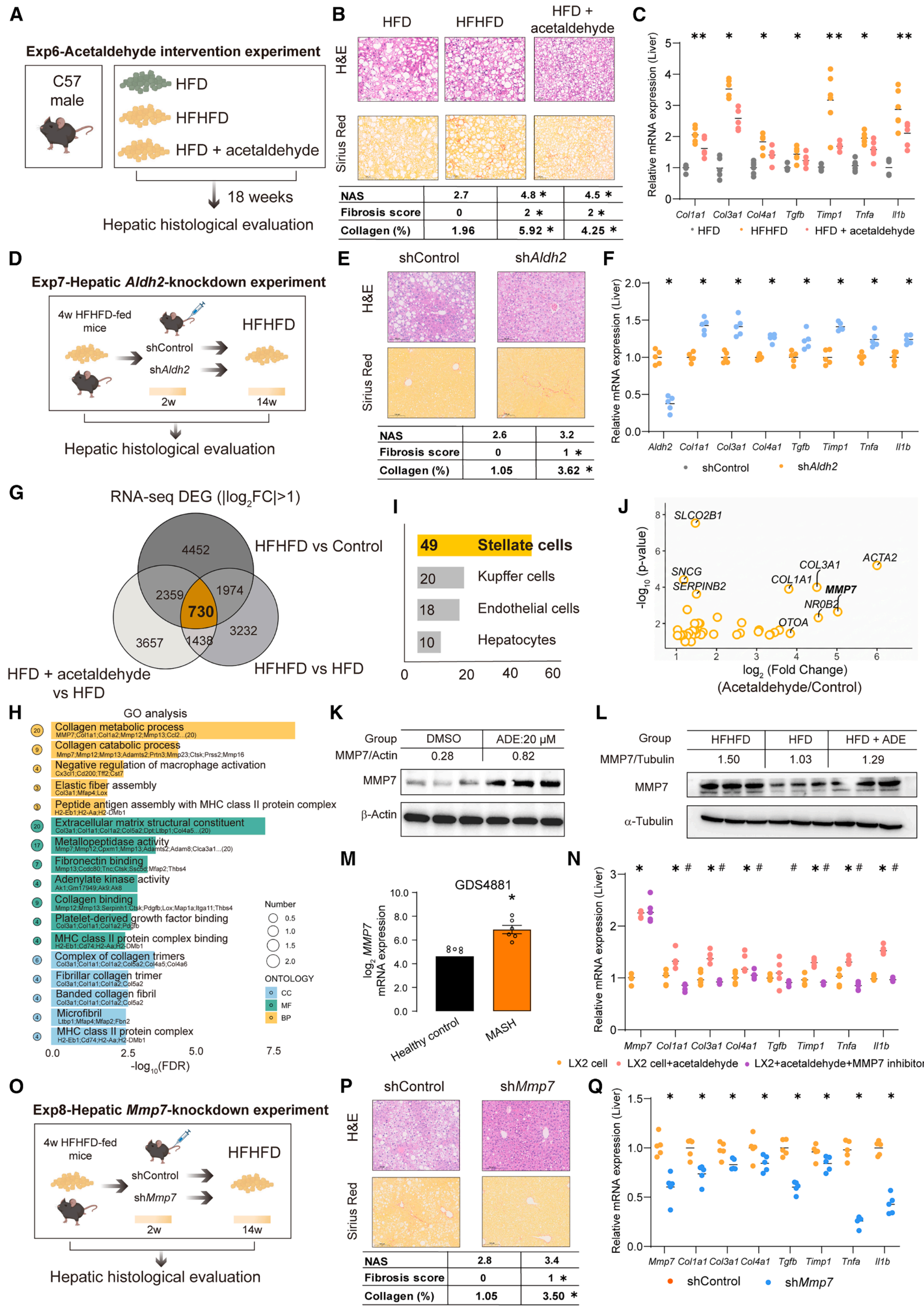
Cell-type mapping analysis indicated that HSCs were predominantly enriched for these transcriptomic alterations, positioning them as central mediators of fibrosis progression (Figure 5I). Supporting this, RNA-seq of LX2 (human HSCs) exposed to low-dose acetaldehyde—reflecting *in vivo* hepatic levels—showed significant upregulation of *MMP7* (Figure 5J). Western blot validation confirmed increased MMP7 expression in LX2 cells treated with acetaldehyde and in liver tissues from acetaldehyde-administered mice (Figures 5K and 5L). Analysis of public human liver transcriptomic datasets (GEO: GDS4881 and GDS4389) revealed elevated *MMP7* levels in patients with MASH and alcoholic hepatitis compared with healthy controls (Figures 5M and S5H). Further qPCR analyses confirmed *MMP7* induction following acetaldehyde exposure, accompanied by heightened pro-fibrotic gene expression (Figure 5N). Importantly, MMP7 inhibition significantly attenuated fibrosis-associated gene expression, confirming its functional role in mediating fibrogenesis (Figure 5N). To definitively establish the causal role of HSCs-derived MMP7 *in vivo*, we employed a cell-type-specific knockdown strategy using an AAV vector driven by the HSCs-specific GfaABC1D promoter. Strikingly, under HFHFD feeding conditions, AAV-sh*Mmp7* mice exhibited significantly ameliorated hepatic fibrosis compared with AAV-shControl mice (Figures 5O–5Q). Furthermore, qPCR analyses revealed elevated *Mmp7* expression in mice with acetaldehyde intervention or AAV-sh*Aldh2* treatment, compared with their respective controls (Figures S5I and S5J), providing additional evidence that acetaldehyde accumulation acts as a potent upstream inducer of HSCs-derived MMP7.

Probiotic strategies targeting acetaldehyde metabolism prevent disease progression

To target gut microbiota-derived endogenous acetaldehyde, we screened probiotics with high acetaldehyde-metabolizing capacity from fecal samples of 51 patients with ALD. These patients had a history of chronic alcohol consumption (>30 years) and exhibited low fibrosis and steatosis scores (FIB-4 < 1.45, NFS < −1.455, and HSI < 36) (Figure 6A). After isolating bacterial clones on selective media, we identified multiple strains, including *Bacteroides uniformis*, *Lacticaseibacillus rhamnosus*, *Parabacteroides distasonis*, *Phocaeicola dorei*, *Bacteroides thetaiotaomicron*, *Klebsiella pneumoniae*, *Lactococcus garvieae*, *Bacteroides cellulosilyticus, Leuconostoc lactis* (*L. lactis*), *Bifidobacterium bifidum* (*B. bifidum*), and *L. salivarius*. Enzymatic screening pinpointed *L. salivarius* as exhibiting superior acetaldehyde clearance, demonstrating 10.9-fold and 5.2-fold greater activity than the standard strain (*L. salivarius* ATCC 11741) under ethanol + acetaldehyde and acetaldehyde-only conditions, respectively (Figure 6B). To definitively validate the taxonomic identity of the isolated bacterial strain, we employed whole-genome sequencing (WGS). Subsequent average nucleotide identity (ANI) analysis and phylogenetic tree construction confirmed that the isolate is a distinct strain of *L. salivarius* (Figures 6C and S6A).

To further enhance its metabolic capacity, we engineered *L. salivarius* to overexpress the microbial bifunctional aldehyde-alcohol dehydrogenase gene (*adhE*) using the pMG36e vector, generating the probiotic strain *L. salivarius* HAM (Figures 6D and S6B). This engineered strain exhibited markedly

(J) Hepatic acetaldehyde levels in HFHFD-fed mice using ADH activity inhibitor 4-MP in animal experiment 4 (5 biological replicates for each group).
(K) Hepatic acetaldehyde levels in whole-body *Aldh2* KO mice compared with wild-type mice in animal experiment 5 (5 biological replicates for each group).
Data are presented as the mean ± SEM. **p* < 0.05 compared with MASLD patients based on the Mann-Whitney U test (B). **p* < 0.05 compared with the HFD group based on the Mann-Whitney U test (C–F). NS, *p* > 0.05 compared with the HFD group based on the Mann-Whitney U test (G–I). NS, *p* > 0.05 compared with the HFHFD group based on the Mann-Whitney U test (J). **p* < 0.05 compared with HFHFD-fed wild-type mice based on the Mann-Whitney U test (K). (A), (J), and (K) were created with BioRender.com.



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increased acetaldehyde clearance—34.9-fold under ethanol + acetaldehyde conditions and 20.6-fold under acetaldehyde alone, compared with the wild-type strain (Figure 6E).

Having confirmed its *in vitro* efficacy, we evaluated the therapeutic potential of *L. salivarius* and *L. salivarius* HAM in a mouse model of diet-induced liver disease. After 16 weeks of HFHFD feeding, mice received oral doses of PBS, standard *L. salivarius* ATCC, isolated *L. salivarius*, or engineered *L. salivarius* HAM every other day for 6 weeks (Figure 6F). Both probiotics significantly improved liver function, evidenced by reductions in ALT and AST levels, decreased liver enlargement and stiffness, and improvements in histological features such as inflammation, fibrosis, and NAS (Figures 6G–6I, S6C, and S6D). Notably, hepatic and fecal acetaldehyde levels were substantially decreased in probiotic-treated mice compared with controls (Figures 6J and 6K). Both strains notably downregulated hepatic *Mmp7* expression and other pro-inflammatory and pro-fibrotic genes, with *L. salivarius* HAM showing superior efficacy (Figures 6L and 6M). The standard *L. salivarius* strain exerted limited effects, primarily modestly reducing inflammation without significantly affecting fibrosis markers (Figure 6L). Long-term administration over 28 weeks similarly ameliorated liver phenotypes, confirming sustained benefits (Figures S7A–S7C). Importantly, metagenomic sequencing revealed that probiotic treatment minimally altered the overall gut microbial diversity (Figures S6E and S6F), indicating targeted activity without disrupting microbiota ecology. Furthermore, we found no statistically significant difference in the abundance of *L. salivarius* between the two groups, suggesting that its specific population size is not the sole driver of the elevated acetaldehyde phenotype observed in MASH patients (Figure S6G).

To rigorously evaluate the *in vivo* biosafety of the probiotic strains, we conducted a subacute toxicity assessment. Mice were treated with a high-dose (2×10^{10} colony-forming units [CFUs] per mouse) of either the wild-type strain (*L. salivarius*) or the engineered strain (HAM) via oral gavage. Histological examination of major organs—including the liver, kidney, heart, spleen, lung, and intestine—via H&E staining revealed no observable pathological changes, tissue necrosis, or inflammatory cell infiltration in either treatment group compared with controls (Figure 7A). Furthermore, serum biochemical analysis demonstrated that key indicators of liver function (ALT, AST, and TBIL) and kidney function (CREA) remained within the normal physiological range, showing no significant alterations following probiotic administration (Figure 7B). Collectively, these data confirm that high-dose administration of *L. salivarius* or *L. salivarius* HAM does not induce systemic toxicity or organ damage, supporting their high safety profile for potential therapeutic application.

DISCUSSION

Excess dietary sugar consumption is epidemiologically associated with metabolic, cardiovascular, and neoplastic diseases,[15] yet its mechanistic contribution to the progression of MASLD-MASH has remained unclear. Our findings reveal that high sugar intake promotes the transition from MASLD to MASH through gut microbial production of acetaldehyde. Importantly, we identified and engineered *L. salivarius* to efficiently metabolize gut luminal acetaldehyde, thereby attenuating liver fibrosis and offering a novel microbiome-targeted therapeutic strategy to prevent MASLD-MASH progression (Figure 7C).

Unlike humans, gut microbiota can ferment sugars such as fructose and glucose into alcohols and aldehydes. Microbial fermentation pathways, including mixed-acid and 2,3-butanediol fermentation, convert pyruvate into various metabolites, notably acetaldehyde and ethanol.[16] For example, *Klebsiella pneumoniae* (*K. pneumoniae*) isolated from patients with auto-brewery syndrome (ABS) can produce substantial ethanol from monosaccharides.[17] Similarly, *Weissella confusa*, *Lactobacillus fermentum*, and *Saccharomyces cerevisiae* ferment sugars into ethanol,[18] while specific lactic acid bacteria can generate acetaldehyde as an intermediate in glucose-to-ethanol conversion.[19]

Figure 5. Acetaldehyde upregulated MMP7 expression in HSCs to drive hepatic fibrosis

(A–C) Animal experiment 6, acetaldehyde intervention experiment (6 biological replicates for each group) (A); representative images of H&E and Sirius Red staining of the liver (scale bars, 100 μm) (B); and relative expression of hepatic pro-inflammatory and pro-fibrotic genes of HFD, HFHFD, and HFD + acetaldehyde groups (C).

(D–F) Animal experiment 7, hepatic *Aldh2*-knockdown experiment (5 biological replicates for each group) (D); representative images of H&E and Sirius Red staining of the liver (scale bars, 100 μm) (E); and relative expression of hepatic pro-inflammatory and pro-fibrotic genes of the shControl group and sh*Aldh2* group (F).

(G) The overlapping DEGs between three pairwise comparisons (HFHFD vs. control in 28-week-old mice from animal experiment 1, HFHFD vs. HFD in 28-week-old mice from animal experiment 1, and HFD vs. HFD + acetaldehyde in animal experiment 6).

(H) GO enrichment analysis using overlapped DEGs.

(I) Cell-type mapping analysis of the 730 overlapped DEGs.

(J) Volcano plot based on the DEGs between acetaldehyde-treated and untreated LX2 (human HSCs).

(K) MMP7 protein expression of LX-2 cells cocultured with low-dose acetaldehyde (ADE) (3 biological replicates for each group).

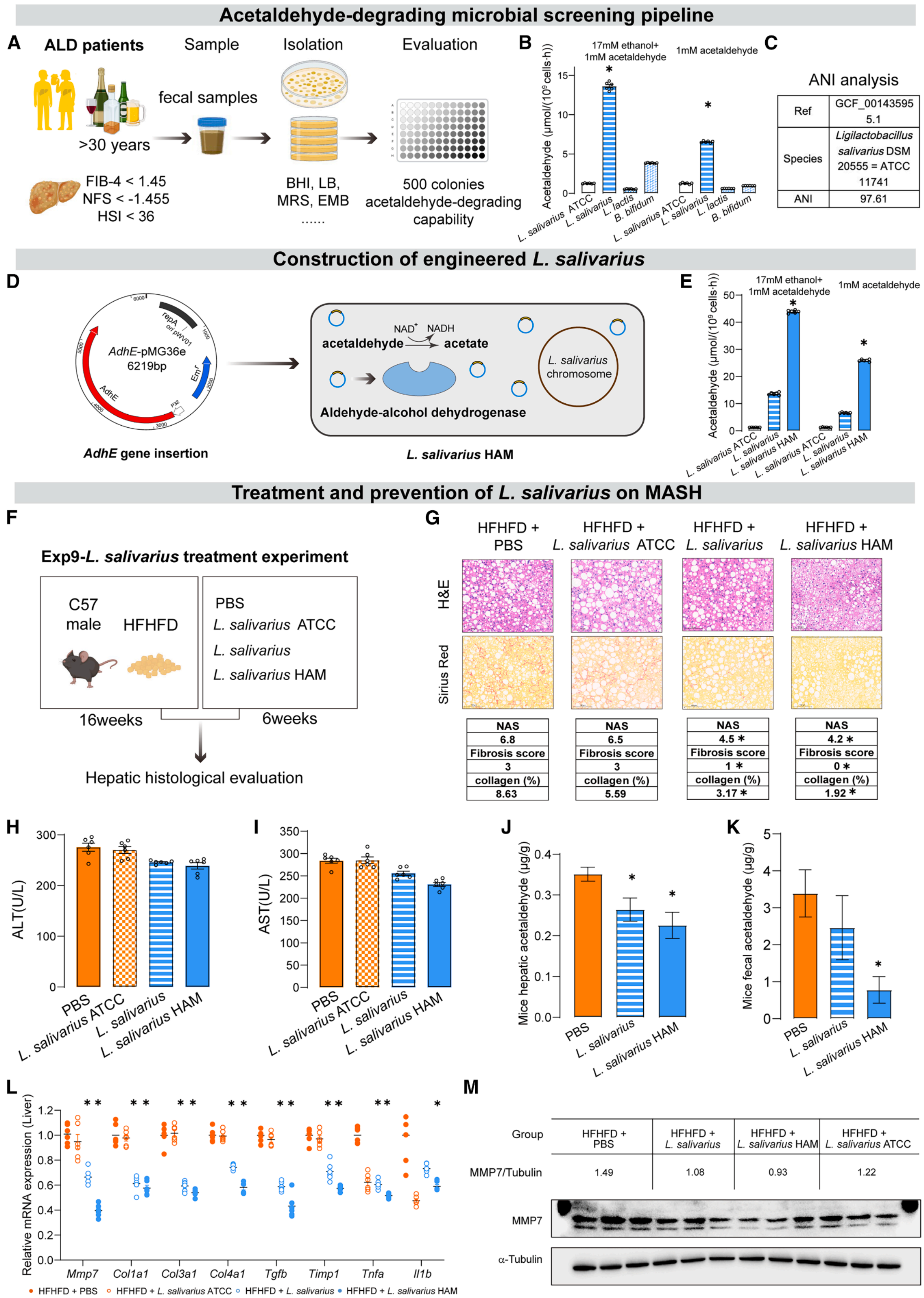
(L) Hepatic MMP7 protein expression of mice administrated low-dose acetaldehyde (ADE) in animal experiment 6 (3 biological replicates for each group).

(M) Relative mRNA expression of *MMP7* in public database (GDS4881).

(N) Relative mRNA expression of *Mmp7*, pro-inflammatory genes, and pro-fibrotic genes of LX2 cell cocultured with acetaldehyde and MMP7 functional inhibitor (5 biological replicates for each group).

(O–Q) Animal experiment 8: hepatic *Mmp7*-knockdown experiment (5 biological replicates for each group) (O), representative images of H&E and Sirius Red staining of the liver (scale bars, 100 μm) (P), and relative mRNA levels of hepatic pro-inflammatory genes and pro-fibrotic genes of the shControl group and sh*Mmp7* groups (Q).

Data are presented as the mean ± SEM. $^{*}p < 0.05$ compared with the HFD group based on the Kruskal-Wallis test with Benjamini-Hochberg adjustment (B and C). $^{*}p < 0.05$ compared with the shControl group based on the Mann-Whitney U test (E and F). $^{*}p < 0.05$ compared with healthy control based on the Mann-Whitney U test (M). $^{*,\#}p < 0.05$ compared with the LX-2 and LX-2 + acetaldehyde groups, respectively, based on the Kruskal-Wallis test with Benjamini-Hochberg adjustment (N). $^{*}p < 0.05$ compared with the shControl group based on the Mann-Whitney U test (P and Q). (A), (D), and (O) were created with BioRender.com.



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Additionally, oral bacteria such as *Streptococcus mutans* can convert glucose into acetaldehyde.[20] Despite these observations, the extent and significance of microbial acetaldehyde production in the gut, especially in the context of liver disease, remain underexplored. Our *in vitro* fecal transformation experiments demonstrate that sugar-responsive gut bacteria collectively contribute to elevated acetaldehyde levels, supporting a microbial role in disease progression. Notably, our clinical analysis revealed a robust positive correlation between fecal acetaldehyde concentrations and fibrosis indices (FIB-4). These findings suggest that gut-derived acetaldehyde holds promise as a noninvasive biomarker for stratifying patients at risk of MASH progression. Future studies with larger cohorts and longitudinal designs are needed to confirm its diagnostic sensitivity, specificity, and utility in clinical practice.

Endogenous aldehydes and alcohols have garnered increasing attention for their pathogenic roles in metabolic diseases. Beyond MASLD-MASH, aldehyde-producing bacteria such as *Klebsiella spp.* have been implicated in ABS.[17] The host's capacity to detoxify these compounds critically influences disease outcomes. Notably, a loss-of-function mutation in ALDH2 (ALDH2*2) significantly elevates endogenous aldehyde levels and predisposes individuals to conditions including Alzheimer's disease, diabetes, cardiovascular disease, and gastrointestinal disorders.[21] Consistent with this, our data show that *Aldh2* knockout mice accumulate hepatic acetaldehyde, which accelerates MASLD-MASH progression. These findings provide a mechanistic basis linking microbial aldehyde production and host detoxification capacity to liver disease pathogenesis, with broader implications for aldehyde-related diseases.

The MMP7 is a zinc- and calcium-dependent endopeptidase with diverse roles in fibrosis.[22,23] Serum MMP7 serves as a biomarker for fibrosis in MASLD patients[24] and contributes to ECM remodeling in kidney fibrosis.[22] MMP7 degrades ECM components such as fibronectin, laminin, and entactin, altering tissue architecture, and promotes fibrogenic signaling by releasing insulin growth factor (IGF)-I/II through cleavage of IGF-binding proteins. Furthermore, MMP7 upregulates collagen genes (e.g., *Col1a2* and *Col3a1*) via pathways including PI3K, p38, extracellular signal-regulated kinase (ERK), Src, and PKA, leading to collagen accumulation in various organs.[25] However, its role in hepatic fibrosis has been less characterized. Our study demonstrates that acetaldehyde induces MMP7 expression in HSCs, driving collagen deposition and accelerating liver fibrosis, thereby elucidating a novel pathogenic mechanism.

While probiotics have shown promise in modulating metabolic and inflammatory pathways, their molecular mechanisms targeting the gut-liver axis in MASH are incompletely understood. Previous studies have employed probiotics to reduce hepatic inflammation and oxidative stress, but their capacity to directly attenuate fibrosis remains underexplored. Here, we present the first evidence that screening and engineering probiotic strains—specifically *L. salivarius*—to enhance acetaldehyde clearance can effectively prevent microbial aldehyde accumulation and mitigate liver fibrosis. This precision microbiome approach underscores the therapeutic potential of targeting microbial metabolites in MASH.

In summary, our study advances understanding of endogenous aldehydes as key mediators in MASLD progression, emphasizing the interplay between diet, gut microbiota, and host detoxification pathways in liver fibrosis. Importantly, our use of engineered probiotics to enhance aldehyde clearance offers a promising avenue for microbiome-based therapies to prevent and treat MASH.

Limitations of the study

While our study establishes a novel link between gut microbial sugar fermentation, aldehydes (specifically acetaldehyde), and hepatic pathology, and demonstrates the therapeutic efficacy of our engineered probiotic, two key limitations warrant mention. First, our metabolomic analysis focused primarily on acetaldehyde as the major pathogenic aldehyde, and a comprehensive screening of the full spectrum of potentially contributory aldehydes produced during microbial sugar fermentation was not performed. Second, although the engineered probiotic showed significant therapeutic effects in preventing hepatic fibrosis, a systematic assessment of its long-term stability, engraftment dynamics, and safety profile within the complex gut environment remains important and is critical for future clinical translation.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources and reagents should be directed to and will be fulfilled by the lead contact, Wei Jia (weijia2@hku.hk).

Figure 6. Probiotic administration halted the MASLD-MASH progression

(A) Schematic illustration of screening criteria of feces from the ALD cohort, probiotics isolation, and acetaldehyde degradation capacity evaluation.

(B) *In vitro* enzymatic assay system to determine the acetaldehyde degradation activity of isolated bacteria (6 biological replicates for each group).

(C) Average nucleotide identity (ANI) analysis of *L. salivarius*.

(D) Schematic illustration of the construction of engineered *L. salivarius* HAM in which the bifunctional aldehyde-alcohol dehydrogenase encoding gene was introduced into screened *L. salivarius*.

(E) *In vitro* enzymatic assay system to determine acetaldehyde degradation activity of *L. salivarius* HAM (6 biological replicates for each group).

(F–L) Animal experiment 9: *L. salivarius* treatment experiment (6 biological replicates for each group) (F); representative images of H&E and Sirius Red staining of the liver (scale bars, 100 μm) (G); serum ALT of mice (H); serum AST of mice (I) of HFHFD + PBS, HFHFD + *L. salivarius* ATCC, HFHFD + *L. salivarius,* and HFHFD + *L. salivarius* HAM groups; hepatic acetaldehyde levels (J) and fecal acetaldehyde levels (K) of HFHFD + PBS; HFHFD + *L. salivarius*, and HFHFD + *L. salivarius* HAM groups; and relative mRNA expression of *Mmp7*, pro-inflammatory and pro-fibrotic genes of HFHFD + PBS, HFHFD + *L. salivarius* ATCC, HFHFD + *L. salivarius*, and HFHFD + *L. salivarius* HAM groups (L).

(M) Protein levels of MMP7 among the HFHFD + PBS, HFHFD + *L. salivarius* ATCC, HFHFD + *L. salivarius*, and HFHFD + *L. salivarius* HAM groups (3 biological replicates for each group).

Data are presented as the mean ± SEM. $^{*}p < 0.05$ compared with the *L. salivarius* ATCC group based on the Kruskal-Wallis test with Benjamini-Hochberg adjustment (B and E). $^{*}p < 0.05$ compared with the HFHFD + PBS group based on the Kruskal-Wallis test with Benjamini-Hochberg adjustment (G–L). (A), (D), and (F) were created with BioRender.com.

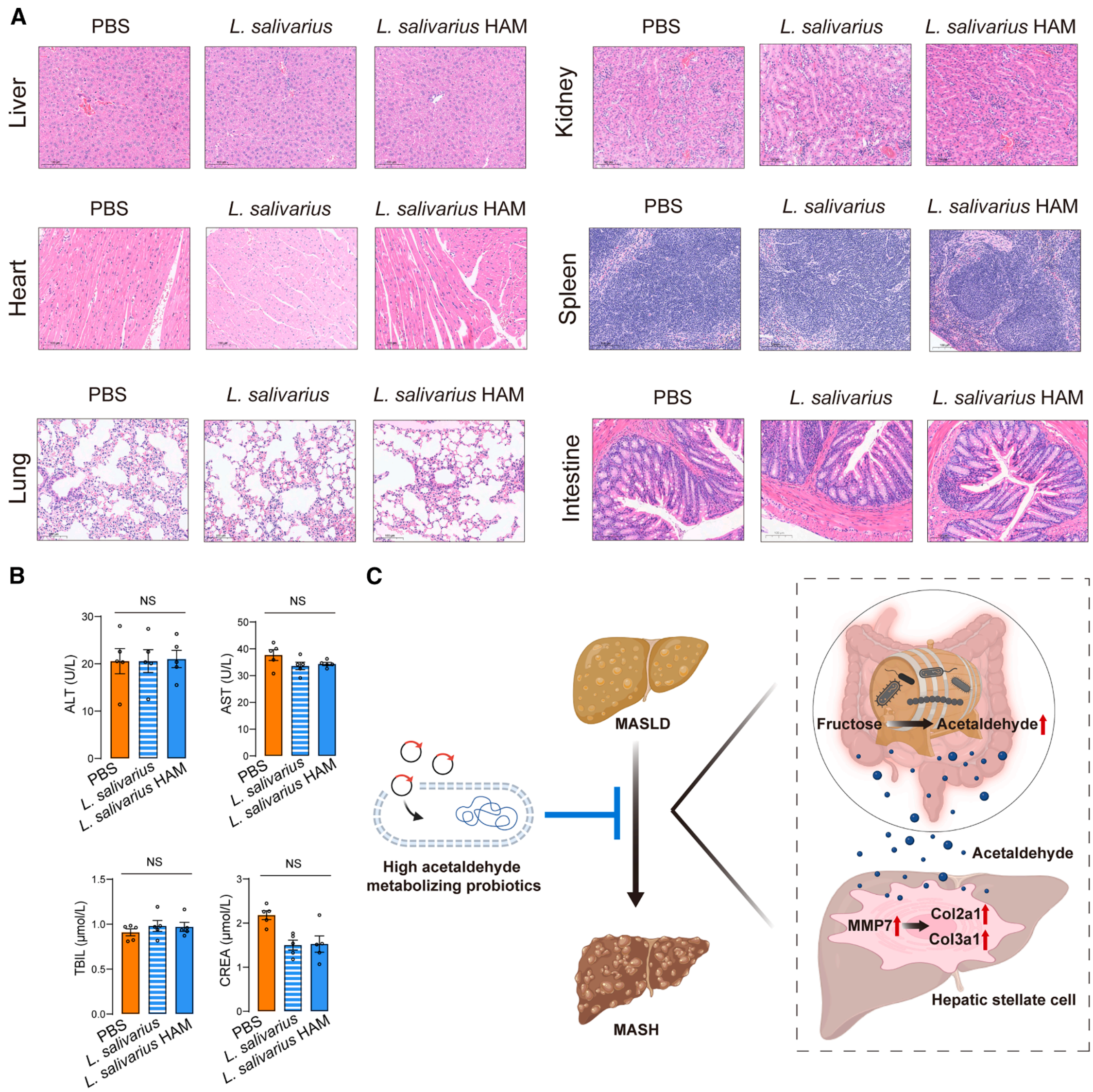

Figure 7. Safety evaluation of *L. salivarius*

(A) Representative images of H&E staining of the main organs of mice in PBS, *L. salivarius*, and *L. salivarius* HAM groups in animal experiment 11 (5 biological replicates for each group); scale bars, 100 μm.

(B) Biochemical analysis (ALT, AST, TBIL, and CREA) of plasma in animal experiment 11 (5 biological replicates for each group).

(C) Microbiota-derived endogenous acetaldehyde upregulates the expression of MMP7 of HSCs, which subsequently increases the collagen synthesis and hepatic fibrosis. Administration of isolated and engineered *L. salivarius* HAM prevented the MASLD-MASH progression through targeting the clearance of endogenous acetaldehyde.

Data are presented as the mean ± SEM. NS, $p > 0.05$ compared with the PBS group based on the Kruskal-Wallis test with Benjamini-Hochberg adjustment (B). (C) was created with BioRender.com.

Materials availability

This study did not generate any new, unique reagents.

Data and code availability

- RNA-seq and metagenomic data of animal experiments have been deposited at GSA and are publicly available as of the date of publication. The accession number of the Bioproject is listed in the key resources table.
- Additional public databases used in this study include UK Biobank (https://www.ukbiobank.ac.uk/), the European NAFLD registry (ENA: PRJEB55534, https://www.ebi.ac.uk/ena/browser/view/PRJEB55534), MetaCyc (https://biocyc.org/download.shtml), and the Gene Expression Omnibus (GEO: GDS4881 and GDS4389, https://www.ncbi.nlm.nih.gov/geo/).
- The scripts used are available in GitHub at https://github.com/zhssakura/ Gut_mcrobiome_2025_codes_SZ.git (v.1.0.0).
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

W.J. was responsible for the concept and design of the study. W.J. and X.Z. organized critical discussions of the results. Y.T. and J.K. drafted the manuscript and conducted experiments and statistical analyses. Y.T., K.D., X.X., and Z.Z. were responsible for generating and analyzing basic clinical data. Y.T., J.K., X.X., Z.Z., Z.R., M.L., Y.L., and F.J. were responsible for animal experiments or cell experiments. C.Y., J.L., and D.Z. were responsible for sample preparation and acetaldehyde analysis. G.J. and X.W. were responsible for providing the essential clinical information. S.Z. was responsible for public metagenomic and metatranscriptomic sequencing collection and annotation. The manuscript was critically reviewed for important intellectual content by T.C., A.Z., and X.Z.

DECLARATION OF INTERESTS

The authors declare no competing interests.

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT OR RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Rabbit polyclonal anti-MMP7	Proteintech	Cat#10374-2-AP; RRID: AB_2144452
Mouse monoclonal anti-alpha-Tubulin	Abcam	Cat#ab7191; RRID: AB_2241126
Rabbit monoclonal anti-β-actin	Cell Signaling Technology	Cat#4970; RRID: AB_2223172
Anti-rabbit, HRP-linked	Cell Signaling Technology	Cat#7074; RRID: AB_2099233
Anti-mouse, HRP-linked	Cell Signaling Technology	Cat#7076; RRID: AB_330924
Bacterial and virus strains		
Leuconostoc lactis	This paper	N/A
Lacticaseibacillus rhamnosus	This paper	N/A
Bacteroides uniformis	This paper	N/A
Parabacteroides distasonis	This paper	N/A
Phocaeicola dorei	This paper	N/A
Bacteroides thetaiotaomicron	This paper	N/A
Bifidobacterium bifidum	This paper	N/A
Ligilactobacillus salivarius	This paper	N/A
Klebsiella pneumoniae	This paper	N/A
Lactococcus garvieae	This paper	N/A
Bacteroides cellulosilyticus	This paper	N/A
Ligilactobacillus salivarius	ATCC	Cat#11741
Ligilactobacillus salivarius HAM	This paper	N/A
Escherichia coli DH5α	Biosharp	Cat#BL1288A
Biological samples		
Feces from humans	This paper	N/A
Serum from mice	This paper	N/A
Liver tissue from mice	This paper	N/A
Intestinal contents from mice	This paper	N/A
Intestinal tissue from mice	This paper	N/A
Chemicals, peptides, and recombinant proteins		
Acetaldehyde	Sigma	Cat#00070; CAS: 75-07-0
Ethanol	Sigma	Cat#493511; CAS: 64-17-5
Vancomycin	Sigma	Cat#1709007
Metronidazole	Sigma	Cat#M1547
Ampicillin	Sigma	Cat#A9393
Neomycin	Sigma	Cat#PHR1491; CAS: 1405-10-3
Phosphate buffered saline	Sigma	Cat#P2272
Paraformaldehyde	Sigma	Cat#158127; CAS: 30525-89-4
RIPA buffer	Beyotime Biotechnology	Cat#P0013C
Glucose	Sigma	Cat#G7021
Fructose	Sigma	Cat#F0127; CAS: 57-48-7
Phenylmethylsulfonyl fluoride	Beyotime Biotechnology	Cat#ST505
4-methylpyrazole	Sigma	Cat#M1387; CAS: 56010-88-9
MMP-7-IN-2	TargetMol	Cat#T78181
Critical commercial assays		
FastPure Cell/Tissue Total RNA Isolation Kit V2	Vazyme Biotech	Cat#RC112-01

(*Continued on next page*)

Continued		
REAGENT OR RESOURCE	SOURCE	IDENTIFIER
PrimerScript RT Reagent Kit	Takara	Cat#RR047Q
Taq Pro Universal SYBR qPCR master mix	Vazyme	Cat#Q712-02
Pierce BCA Protein Assay Kit	Thermo Fisher	23227
Clarity Western ECL Substrate	Bio-Rad	Cat#1705060
Deposited data		
The raw data of RNA-seq generated	This paper	CRA061281
The raw data of metagenomic gene sequencing	This paper	CRA061557
Experimental models: Cell lines		
LX-2	Sigma	SCC064
Experimental models: Organisms/strains		
C57BL/6J mouse	Shanghai SLAC Laboratory Animal Co., Ltd	N/A
sh*Aldh2* C57BL/6J mice	This paper	N/A
sh*Mmp7* C57BL/6J mice	This paper	N/A
$Aldh2^{-/-}$ mice	N/A	N/A
Software and algorithms		
GraphPad Prism 8 software (GraphPad)	GraphPad Software	http://www.graphpad.com/
Adobe Illustrator	Adobe	https://www.adobe.com/products/illustrator.html
TargetLynx 4.2	Waters Corporation	N/A
ImageJ	National Institutes of Health (NIH)	https://imagej.nih.gov/ij/
SPSS 26.0	IBM SPSS	https://www.ibm.com/hk-en/products/spss-statistics
R studio	RStudio	https://www.r-project.org/
Microsoft 365	Microsoft	https://www.microsoft.com/zh-cn/microsoft-365/buy/compare-all-microsoft-365-products?tab=1
Other		
Rodent diet with 60 kcal% fat	Research Diets	Cat#D12492
Rodent diet with 40 kcal% fat, 20 kcal% fructose and 2% cholesterol	Research Diets	Cat#D09100310
Rodent Diet with 10 kcal% Fat	Research Diets	Cat# D12450J
PAGE Gel Fast Preparation Kit	EpiZyme	Cat#PG112
PageRuler Prestained Protein Ladder	Thermo Fisher	Cat#26616
Immobilon-PSQ PVDF Membrane	Millipore	Cat#ISEQ00010
Fetal Bovine Serum	Gibco	Cat#10100147C
Penicillin-streptomycin (5000 U/mL)	Gibco	Cat#15140122
DMEM	Invitrogen	Cat#11965092
M9 minimal salts	Thermo Fisher	Cat#A1374401
MacConkey agar	Thermo Fisher	Cat#CM0007B
Brain Heart Infusion (BHI) agar	Thermo Fisher	Cat#CM1136B
De Man, Rogosa and Sharpe (MRS) agar	Thermo Fisher	Cat#CM1153B
Eosin Methylene Blue (EMB) agar	Thermo Fisher	Cat#CM0069B

EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

Human study

All the human studies were approved by the Ethics Committee of Shanghai Sixth People's Hospital affiliated to Shanghai Jiao Tong University School of Medicine Hospital in accordance with the World Medical Association's Declaration of Helsinki. The human fecal specimens in the Biorepository were derived from multiple large cohort studies of metabolic diseases, including T2DM and MASLD.

Human cohort 1

The UK Biobank is a large-scale biomedical database that includes comprehensive health and genetic information from over 500,000 participants, all of whom are aged between 40 and 69 years at recruitment. For the present analysis, a subset of the UK Biobank data was utilized, consisting of 210,861 participants with complete dietary intake data. Liver disease-related data were obtained based on ICD-10 coding (C22- malignant neoplasm of liver and intrahepatic bile ducts, K72- hepatic failure, not elsewhere classified, K73- chronic hepatitis, not elsewhere classified, K74- fibrosis and cirrhosis of liver, K76.0- fatty (change of) liver, not elsewhere classified) for liver diseases diagnoses. The liver disease group included 5,904 individuals, while the healthy control group consisted of 26,723 participants, matched for age, sex, and other key demographic variables (Table S1).

Human cohort 2

It included 146 individuals from the European NAFLD registry, for whom raw metagenomic sequencing data of fecal samples were downloaded.[11] This cohort consisted of 20 MASLD, 50 MASH-0 (inflammatory stage), and 76 MASH-1 (fibrosis stage) patients (Table S2A). Each participant in this cohort underwent liver biopsy for clinical diagnosis of their respective condition, with final diagnoses confirmed based on histological assessment.

The grouping information, including the classification of individuals into healthy, MASLD, and MASH categories, was obtained directly from the European NAFLD registry database.

Human cohort 3

It consisted of 25 patients diagnosed with MASLD (Table S3). Based on NFS, FIB-4 scores and HSI, the cohort was stratified into three groups: $HSI^-FIB\text{-}4^-$ (n=5), $HSI^+FIB\text{-}4^-$ (n=8), and $HSI^+FIB\text{-}4^+$ (n=12).[12–14] The stratification was performed using specific cutoff values for both the HSI and FIB-4 scores, as outlined below.

$HSI^-FIB\text{-}4^-$ Group (n=5): This group included individuals who had both an HSI score of < 36 and a FIB-4 score of < 1.45. Patients in this group had a low likelihood of both hepatic steatosis and advanced fibrosis. $HSI^+FIB\text{-}4^-$ Group (n=8): The $HSI^+FIB\text{-}4^-$ group consisted of individuals with an HSI score of ≥ 36, and a FIB-4 score of < 1.45, suggesting no significant fibrosis. This group had hepatic steatosis but a low likelihood of advanced fibrosis. $HSI^+FIB\text{-}4^+$ Group (n=12): The $HSI^+FIB\text{-}4^+$ group included individuals who had both an HSI score of ≥ 36 and a FIB-4 score of ≥ 1.45. This group represents patients with hepatic steatosis as well as a high likelihood of advanced fibrosis or cirrhosis.

HSI Scoring System:

$$HSI = 8 \times (ALT/AST) + BMI + 2\ (if\ type\ 2\ diabetes)\ + 2\ (if\ female)$$

FIB-4 Scoring System:

$$FIB - 4 = \frac{Age \times AST}{Platelets \times \sqrt{ALT}}$$

Criteria for MASLD

The diagnosis is based on the following criteria: (1) Evidence of hepatic steatosis through imaging (e.g., ultrasound, CT, MRI) or liver biopsy; (2) Presence of at least one metabolic risk factor, such as obesity (BMI ≥ 25 kg/m^2), abdominal obesity, high blood glucose, diabetes, hypertension, dyslipidemia (e.g., low HDL-C or high triglycerides), or use of lipid-lowering medication; (3) Exclusion of other liver diseases, such as alcohol-related liver disease, drug-induced liver injury, or viral hepatitis; and (4) Classification of the disease stage, ranging from simple steatosis to steatohepatitis and advanced fibrosis or cirrhosis. MASLD is differentiated from other forms of liver disease by its association with metabolic dysfunction and the exclusion of alcohol as a causative factor.

Human cohort 4

It consisted of 51 patients diagnosed with ALD, all with a history of more than 30 years of alcohol consumption. These patients were diagnosed following the latest clinical guidelines for ALD.[26]

We used non-invasive fibrosis and steatosis scoring systems. Specifically, patients were included if they met the following criteria: FIB-4 score < 1.45, NFS < -1.455, HSI < 36[12-14].

Criteria for ALD

ALD[26] is diagnosed based on a combination of chronic alcohol consumption history, clinical symptoms, and exclusion of other potential causes of liver disease. The primary diagnostic criteria include: (1) Chronic alcohol consumption: This is defined as more than 30 grams of alcohol per day for men and more than 20 grams per day for women over a period of at least 5 years. (2) Exclusion of other liver diseases: Other potential causes of liver injury, such as viral hepatitis (e.g., hepatitis B and C), MASLD, autoimmune liver diseases, and drug-induced liver injury, must be excluded through laboratory tests, imaging, and, when necessary, liver biopsy. (3) Clinical evidence of liver damage: This includes elevated liver enzymes (AST and ALT), hepatomegaly, jaundice, and histological features such as alcoholic steatosis, alcoholic hepatitis, and cirrhosis observed on liver biopsy (if available).

Mouse models

All animal studies were approved by the Institutional Animal Care and Use Committee at the Center for Laboratory Animals, Shanghai Sixth People's Hospital Affiliated to Shanghai Jiao Tong University School of Medicine (Shanghai, China). All the mice were housed in a specific-pathogen-free (SPF) facility under controlled conditions (20-22°C, 45 ± 5% humidity, 12 h light/dark cycle) with free access to food and water. After a one-week acclimation period on a standard chow diet (TP23522, Trophic Animal Feed High-tech Co., Ltd, China), the mice were fed either a chow diet, high-fat diet (HFD; D12492, Research Diets, Inc), or high-fat high-fructose diet (HFHFD; D09100310, Research Diets, Inc) for the experiments. Body weight and food intake were recorded weekly throughout the study.

METHOD DETAILS

Animal experiments

Animal experiment 1: Long-term HFD/HFHFD model

Mice (6-8 weeks old C57BL/6J, male) were randomly divided into three groups treated with different diets. (1) HFD group (n=6): mice were fed with HFD; (2) HFHFD group (n=6): mice were fed with HFHFD; (3) Control group (n=6): mice were fed with a standard chow. Six mice from three groups were euthanized after 18-week, 28-week, 56-week. The liver specimens were collected for RNA sequencing and histological evaluations. Intestinal tissues were collected for mRNA expression quantification and the detection of acetaldehyde and ethanol. Portal vein serum was harvested for the detection of acetaldehyde and ethanol.

Animal experiment 2: HFD coupled with fructose-supplemented drinking water experiment

Mice (6-8 weeks old C57BL/6J, male) were randomly divided into two groups treated with different diets. (1) HFD group (n=5): mice were fed with HFD; (2) HFD + 10%fructose group (n=5): mice were fed with HFD coupled with 10% fructose water. Five mice from each group were euthanized after 28 weeks. The liver specimens were collected for further histological evaluations.

Animal experiment 3: HFHFD + ABX experiment

Mice (6-8 weeks old C57BL/6J, male) were divided into two groups: (1) HFHFD group (n=6): mice were fed with HFHFD coupled with sterile water; (2) HFHFD + ABX group (n=6): mice were fed with HFHFD coupled with ABX (including vancomycin: 100 mg/kg/day, neomycin: 200 mg/kg/day, metronidazole: 200 mg/kg/day, ampicillin: 200 mg/kg/day) in their drinking water *ad libitum*. Mice were euthanized after 18-weeks intervention. The liver specimen was collected for mRNA expression quantification and histological evaluations.

Animal experiment 4: ADH activity inhibition experiment

Mice (6-8 weeks old C57BL/6J, male) were divided into two groups: (1) HFHFD group (n=5): mice were fed with HFHFD for 12 weeks; (2) HFHFD + 4-MP group (n=5): mice were fed with HFHFD for 11 weeks coupled with one-week 4-MP (25mg/kg/day) by intraperitoneal injection. Mice were euthanized after one week of intervention. The liver specimens were collected for the detection of acetaldehyde and ethanol.

Animal experiment 5: MASH model in Aldh2*-KO mice*

Mice (6-8 weeks old C57BL/6J, male) were divided into two groups: (1) Wild type + HFHFD group (n=5): mice were fed with HFHFD; (2) ALDH2$^{-/-}$ + HFHFD group (n=5): ALDH2$^{-/-}$ mice were fed with HFHFD. Mice were euthanized after 18-weeks. The liver specimens were collected for histological evaluations.

Animal experiment 6: Acetaldehyde intervention experiment

Mice (6-8 weeks old C57BL/6J, male) were divided into three groups: (1) HFD group (n=6): mice were fed an HFD and provided with regular drinking water *ad libitum*; (2) HFHFD group (n=6): mice were fed with HFHFD and provided with regular drinking water *ad libitum*; (3) HFD + acetaldehyde group (n=6): mice were fed with HFD coupled with acetaldehyde in the drinking water at a dose of 50 mg/kg/day. Mice were euthanized after 18-weeks intervention. And the liver specimens were collected for RNA-sequencing, mRNA expression quantification, western blot, and histological evaluations.

Animal experiment 7: Hepatic Aldh2*-knockdown experiment*

A total of 10 mice (6-8 weeks old C57BL/6J, male) were fed with HFHFD for 4 weeks. Then, each 5 were intravenously injected with AAV-shControl (1x10^{12}vg/mL) and AAV-sh*Aldh2* (1x10^{12}vg/mL) for 2 weeks to knock down the hepatic *Aldh2*. After that, mice were fed HFHFD for another 14 weeks. The liver specimens were collected for histological evaluations and mRNA expression quantification.

Animal experiment 8: Hepatic Mmp7*-knockdown experiment*

A total of 10 mice (6-8 weeks old C57BL/6J, male) were fed with HFHFD for 4 weeks. Then, each 5 were intravenously injected with AAV-sh*Control* (1x10^{12}vg/mL) and AAV-sh*Mmp7* (1x10^{12}vg/mL) for 2 weeks to knock down the hepatic *Mmp7*. After that, mice were fed HFHFD for another 14 weeks. The liver specimens were collected for histological evaluations and mRNA expression quantification.

Animal experiment 9: L. salivarius *treatment experiment*

All mice (6-8 week old C57BL/6J, male) were fed with HFHFD for 16 weeks and then divided into four groups (n=6 per group): (1) HFHFD group: mice were fed with HFHFD and gavaged with 200 μL sterilized PBS per mouse every other day; (2) Live *L. salivarius* ATCC group: mice were treated with HFHFD and gavaged with 200 μL live *L. salivarius* ATCC (resuspended in sterile PBS at 2x10^9 CFU/mL) per mouse every other day; (3) Live *L. salivarius* group: mice were treated with HFHFD and gavaged with 200 μL live *L. salivarius* (resuspended in sterile PBS at 2x10^9 CFU/mL) per mouse every other day; (4) Live *L. salivarius* HAM group: mice were treated with HFHFD and gavaged with 200 μL live *L. salivarius* HAM (resuspended in sterile PBS at 2x10^9 CFU/mL) per mouse every other day. The mice were euthanized after 6 weeks of colonization, and liver specimens were collected for mRNA expression quantification, western blot, the detection of acetaldehyde and ethanol, and histological evaluations. Serum samples were collected for biochemical analysis to assess liver function.

Animal experiment 10:* L. salivarius *prevention experiment

Mice (6-8 week old C57BL/6J, male) were divided into three groups (n=5 per group): (1) HFHFD group: mice were fed with HFHFD and gavaged with 200 μL sterilized PBS per mouse every other day; (2) Live *L. salivarius* group: mice were treated with HFHFD and gavaged with 200 μL live *L. salivarius* (resuspended in sterile PBS at 2x10^9 CFU/mL) per mouse every other day; (3) Live *L. salivarius* HAM group: mice were treated with HFHFD and gavaged with 200 μL live *L. salivarius* HAM (resuspended in sterile PBS at 2x10^9 CFU/mL) per mouse every other day. The mice were euthanized after 28 weeks of colonization, and liver specimens were collected for evaluation using qPCR, western blot, and histological features. The serum samples were collected for mRNA expression quantification, and histological evaluations.

***Animal experiment 11: Safety evaluation of* L. salivarius**

Mice (6-8 week old C57BL/6J, male) were divided into three groups (n=5 per group): (1) Control group: mice were fed with control diet and gavaged with 200 μL sterilized PBS per mouse daily; (2) *L. salivarius* group: mice were treated with control diet and gavaged with 200 μL live *L. salivarius* (resuspended in sterile PBS at 1x10^{11} CFU/mL) per mouse daily; (3) *L. salivarius* HAM group: mice were treated with control diet and gavaged with 200 μL live *L. salivarius* (resuspended in sterile PBS at 1x10^{11} CFU/mL) per mouse daily. The mice were euthanized after 28 days of colonization. Serum was harvested for biochemical analysis. The liver, kidney, heart, spleen, lung and intestine specimens were collected for safety evaluation.

UK Biobank data analyses

Based on International Classification of Diseases, 10^{th} Revision (ICD-10) codes, participants with liver diseases were identified, including C22, K72, K73, K74, K76.0. The daily total sugar intake composed of sucrose, glucose, and fructose were utilized to perform the generalized additive model (GAM) and Kaplan-Meier (KM) survival analyses between the liver disease group (n = 5,904) and the healthy control group (n = 26,723). GAMs were implemented to examine nonlinear relationships between the daily total sugar intake composed of sucrose, glucose, and fructose and the possibility of death or the probability of liver disease, while adjusting for potential confounders including age, sex, BMI and smoking status. The GAM framework allows for flexible modeling of smooth, non-parametric trends while controlling for covariates. Model fit was assessed using diagnostic plots, including residual checks and smooth term evaluations. Statistical significance was determined at *$p < 0.05$, and all analyses were performed using R software (version 4.5.1). KM survival analyses were performed to assess the time-to-event outcomes between different groups. Survival probabilities were estimated using the non-parametric KM method, which accounts for censored data. Log-rank tests were used to compare survival distributions across groups, with statistical significance set at *$p < 0.05$. Covariate-adjusted survival analyses were conducted using Cox proportional hazards models to control for potential confounders, including age, sex, BMI, and smoking status. The proportional hazards assumption was verified using Schoenfeld residuals.

Biochemical analysis

For animal experiments, serum ALT, AST, TBIL and CREA biochemical parameters were measured using a TBA-40FR Fully Automatic Biochemical Analyzer (TOSHIBA, Japan) following the manufacturer's protocol.

Histopathological examination

Liver tissue samples were resected, fixed in 4% paraformaldehyde, and embedded in paraffin following standard protocols. Paraffin-embedded samples were prepared and subsequently stained with H&E staining and Sirius Red staining for histological analysis. Imaging was performed using an ECLIPSE TS2R inverted fluorescence microscope (Nikon, Tokyo, Japan). Hepatic steatosis was quantified by evaluating the percentage of fat droplet-containing hepatocytes, with scoring performed according to established criteria from published studies.[27]

Fecal transformation experiment

Fecal samples were dissolved in phosphate-buffered saline (PBS) to a fixed volume of 1 mL under anaerobic conditions, followed by two washing cycles with PBS via centrifugation (4,000 × *g*, 10 min). A 200 μL aliquot of the washed fecal suspension was then incubated with 45 μL of 1 M fructose or glucose, 90 μL of 5 × M9 minimal medium, and 115 μL deionized water, yielding a final reaction volume of 450 μL. After 20 hours of incubation at 37°C, the mixture was centrifuged (12,000 × *g*, 10 min, 4°C). The supernatant was collected and stored at -80°C for subsequent acetaldehyde and ethanol quantification by UPLC-UV and GC-MS, respectively.

Acetaldehyde quantification by UPLC-UV

In this study, acetaldehyde was quantified using UPLC-UV following derivatization with 2,4-dinitrophenylhydrazine (DNPH). Bacterial supernatants, as well as liver and fecal samples, were homogenized in 150 μL of extraction solvent (acetonitrile: water, 1:1, v/v) for 5 min and subsequently centrifuged at 12,700 rpm for 10 min at 4 °C. The pH of the resulting supernatant (60 μL) was adjusted by adding 7.5 μL of 3 M sodium phosphate buffer. Derivatization was carried out by adding 7.5 μL of DNPH solution (2 mg/mL in 6 M HCl) and incubating the mixture at 60 °C for 20 min. After a second centrifugation step, the supernatant was collected for UPLC-UV analysis.

Chromatographic separation was performed on a CORTECS UPLC C18 column (1.6 μm, 2.1 x 100 mm) maintained at 45 °C. The mobile phase consisted of water (A) and acetonitrile (B). A gradient elution program at a flow rate of 0.4 mL/min was employed as follows: 0-8 min (30% B), 8-10 min (30-65% B), 10-10.1 min (65-100% B), 10.1-12.5 min (100% B), 12.5-12.6 min (100-30% B), and 12.6-15 min (30% B). A 5 μL aliquot of each sample was injected for analysis. The PDA detector was set at 365 nm.

Ethanol quantification by GC-MS

Ethanol was quantified using Gas Chromatography-Mass Spectrometry (GC-MS), employing a Waters Xevo TQ-S triple quadrupole mass spectrometer (Waters Corp., USA) coupled to an Agilent 7890B gas chromatograph (Agilent Technologies, USA). Bacterial supernatants, liver tissues, and fecal samples were extracted with 150 μL of solvent (acetonitrile: water, 1:1, v/v) for 5 min. Separation was achieved on a DB-WAX UI capillary column (30 m × 0.25 mm i.d., 0.25 μm film thickness; Agilent Technologies) with the following temperature program: initial temperature 35 °C (hold 2 min), raised to 110 °C at 15°C/min, then to 220 °C at 35 °C/min (hold 2 min), resulting in a total run time of 12.14 min. The GC system was operated in split mode (50:1 ratio) with an injection volume of 1 μL. The injector, ion source, and transfer line temperatures were maintained at 220 °C, 200 °C, and 220 °C, respectively. The carrier gas (helium) flow rate was maintained at 1.0 mL/min. MS detection was performed in single ion recording (SIR) mode, monitoring m/z 45 for ethanol (retention time: 3.69 min).

Metagenomic sequencing

Metagenome-assembled genomes (MAGs)

Microbial genomic DNA was extracted from mice cecal contents using the DNeasy PowerSoil tool kit (QIAGEN, Valencia, CA, USA), with quantity and quality assessed by NanoDrop ND-1000 spectrophotometer (Thermo Fisher Scientific, Waltham, MA, USA) and agarose gel electrophoresis. Quality-approved DNA samples were used to prepare metagenomic shotgun libraries with Illumina TruSeq Nano DNA LT Library Preparation Kit. The PE150 strategy of each library has been sequenced on the Illumina HiSeq X-10 platform (Illumina, San Diego, CA, USA) through Personal Biotechnology Co., LTD. (Shanghai, China). To understand the key roles of human gut microbiome in the progression from MASLD to MASH, we downloaded 146 metagenomic datasets of fecal samples from the European NAFLD registry (Project accession number: PRJEB55534), including the metadata for liver biopsy-confirmed diagnoses. The cohort comprised 20 MASLD, 50 MASH-0 (inflammatory stage) and 76 MASH-1 (fibrosis stage) patients. More details in Human cohort 2.

Metagenomic sequences obtained from ten mouse fecal samples (see details in Animal experiment 1: Long-term HFD/HFHFD model), as well as 146 human gut microbiomes (see details in Human cohort 2, Table S2A) were analyzed to reconstruct MAGs for comparative analysis. In brief, paired-end reads (2 × 150 bp) were quality filtered and trimmed with Trimmomatic v.0.39 using the following parameters: "TruSeq3-PE-2.fa:2:30:10:6:true LEADING:20 TRAILING:20 HEADCROP:5 SLIDINGWINDOW:4:20 MINLEN:50"[28]. Filtered sequencing reads were *de novo* assembled using SPAdes v4.2.0 in meta-mode, with the kmer size ranging from 21 to 101 bp and an interval of 20 bp.[29] Metagenome binning was performed using CONCOCT, MaxBin2 and metaBAT2, imbedded under the pipeline MetaWRAP v1.3.2,[30] with minimum contig length of 1,000 bp to MAGs generated by CONCOCT and MaxBin2, and minimum contig length of 1,500 bp to bin generated by metaBAT2. The binning results from the above approaches were then used as inputs for genomic binning refinement consolidated by MetaWRAP v1.3.2, which refined bins with completeness of greater than 50% and contamination of less than 5%, assessed by CheckM based on the presence of single-copy genes. A total of 5,020 refined MAGs were obtained (see in Table S2C).

Taxonomic classification and dereplication

Taxonomy was assigned to the refined MAGs based on the Genome Taxonomy Database (GTDB) release 226[31] using GTDB-Tk v2.4.1 (see in Table S2C).[32] The relative abundances of MAGs were calculated for each sample based on metagenomic reads mapped back to MAGs using CoverM v0.7.0.[33] Completeness, contamination, and heterogeneity of MAGs were assessed using ChechM2 v1.1.0 based on machine learning approach, and 4,912 refined MAGs passed quality assessment (with completeness of greater than 50% and contamination of less than 5%, see in Table S2D).[34] These refined MAGs were then clustered using dRep v3.6.2 at a 97% average nucleotide identity (ANI) threshold, and the representative MAGs (rMAGs) were obtained from each dRep cluster with the highest completeness and the lowest contamination.[35] A total of 1,187 rMAGs, including 716 high-quality MAGs (with completeness of 97.6 ± 2.9% and contamination of 0.7 ± 1.0%) and 471 medium-quality MAGs (with completeness of 73.6 ± 11.0% and contamination of 2.1 ± 1.9%) were retained for subsequent analyses (see in Table S2E).[36]

Functional annotation and metabolic pathway prediction of rMAGs

Functional annotation and MetaCyc pathway prediction for 1,187 rMAGs were conducted following published methods.[37] In brief, enzymes were identified by performing TBLASTN using the 'gapseq find' module of gapseq v.1.4.0 (using bit-score and coverage cut-offs of 100 and 70%, respectively) on the protein sequences in the UniProt Knowledgebase (UniRef version 2024_11) against the nucleotide sequences of each rMAG.[38] The "unreviewed" reference sequences from the UniProt database (UniRef50 clusters) were also considered if no TBLASTN hits were found for the "reviewed" ones (UniRef90 clusters). The presence of metabolic pathways was identified based on pathway completeness, which was estimated by the presence of relevant enzymes. A pathway was considered present when: 1) pathway completeness was $\geq$ 80% if no "key enzyme" were detected; 2) $\geq$ 66.7% if all key enzymes were detected; and 3) $\geq$ 33.3% if not all enzymes in the pathway are included in the UniProt Knowledgebase.[38]

A matrix showing the presence (1) or absence (0) of MetaCyc metabolic pathways across 4,912 refined MAGs was generated by mapping metabolism of each rMAG to the remaining MAGs within the same dRep cluster. Additionally, a matrix including the relative abundance of each microbial species (identified at 97% ANI distance[39]) across the 146 human and ten mouse gut microbiomes was obtained. Finally, the percentages of metabolic pathways present in each gut microbiome sample were estimated based on the relative abundance along with the metabolic pathway prediction results for the gut microorganisms, which were used for subsequent analyses.

Metatranscriptomic sequencing

Total RNA was extracted from human fecal samples using the RNA PowerSoil Total RNA Isolation Kit (MoBio, USA) according to the manufacturer's instructions. RNA concentration and quality were assessed using a UV spectrophotometer and 1.5% agarose gel electrophoresis, while RNA integrity (RIN $\geq$ 5.5) was confirmed using an Agilent 2100 Bioanalyzer (Agilent, USA). Following the removal of ribosomal RNA (rRNA), sequencing libraries were constructed using the TruSeq Stranded mRNA LT Sample Prep Kit (Illumina, USA). High-throughput sequencing was performed on the Illumina NovaSeq platform (Personal Biotechnology Co., Ltd., Shanghai, China) using a paired-end 150 bp (PE150) strategy.

For data analysis, raw reads were processed to remove adapters using Cutadapt (v1.17), and low-quality bases were trimmed using fastp (v0.20.0). Host contamination was removed by aligning reads to the human genome using BMTagger, and residual rRNA was filtered using SortMeRNA (v4.2.0). Clean reads were assembled de novo using Trinity (v2.11.0), and the resulting contigs were clustered using MMseqs2 to generate a non-redundant gene catalog. Gene abundance was quantified using Salmon in quasi-mapping mode to calculate Transcripts Per Million (TPM).

Whole genome sequencing (WGS) of *L. salivarius*

Genomic DNA was extracted from the bacterial isolate using the CTAB method, and its concentration, quality, and integrity were assessed using a Qubit Fluorometer and a NanoDrop Spectrophotometer. For genome sequencing, a hybrid approach combining short-read and long-read platforms was employed to ensure high accuracy and continuity. Sequencing libraries were constructed using the TruSeq DNA Sample Preparation Kit (Illumina, USA) for short-read sequencing and the Template Prep Kit (Pacific Biosciences, USA) with BluePippin size selection for long-read sequencing. The sequencing was performed on the Illumina NovaSeq platform and the PacBio Sequel platform (Personal Biotechnology Co., Ltd., Shanghai, China).

Data assembly was performed after adapter contamination removal and data filtering using AdapterRemoval and SOAPec. The filtered short reads were assembled by SPAdes and A5-miseq, while long reads were assembled using Flye and Unicycler to construct scaffolds and contigs. The final genome sequence was acquired after rectification using Pilon software. To definitively confirm the taxonomic identity of the isolate, ANI analysis was performed to quantify the genomic similarity between the isolate and reference *Ligilactobacillus salivarius* strains. Furthermore, a phylogenetic tree was constructed based on the whole-genome sequence to visualize the evolutionary clustering and determine the precise phylogenetic position of the isolated strain.

RNA sequencing

Total RNA was extracted using triazol reagent (Invitrogen Life Technologies) and qualified using a NanoDrop spectrophotometer (Thermo Scientific). From 3 micrograms total RNA, mRNA was enriched using poly T oligo attached magnetic beads and fragmented under high temperature with divalent cations. The first strand cDNA synthesis was performed with random oligonucleotide and Super Script II reverse transcriptase, followed by second strand cDNA synthesis using DNA polymerase I and RNase H. The remaining overhang was blunt-ended using ectomodulase/polymerase, followed by enzyme removal. After 3' adenylation, Illumina adapters were ligated, and 400-500bp fragments were size-selected using AMPure XP system (Beckman Coulter, Beverly, CA, USA). Library amplification (15 cycles) was performed with Illumina PCR primers Cocktail, followed by purification and quality assessment on a Bioanalyzer 2100 system (Agilent). Finally, paired-end sequencing was conducted on an Illumina NovaSeq 6000 platform at Personal Biotechnology Co. (Shanghai).

Cell-type mapping analyses

A list of 730 differentially expressed genes (DEGs) were identified from the bulk RNA-seq data. The MSigDB C8 "Cell Type" collection via the msigdbr R package was leveraged to identify the liver cell type that these DEGs most strongly map onto. All human C8 gene sets (msigdbr(species="Homo sapiens", category="C8")) were downloaded and both gene symbols and gene-set names were converted to upper case. Then, gene sets were filtered to retain only those whose names included "LIVER", yielding exclusively those curated from liver-derived cell clusters. The DEG list was intersected with each liver gene set, and the number of overlapping DEGs in each set was counted. Individual gene-set labels (e.g. "C21_STELLATE_CELLS_1", "C33_STELLATE_CELLS_2") were collapsed into broader categories-"Stellate cells", "Kupffer cells", "Endothelial cells", "Hepatocytes", and "Other"-by simple keyword matching. Subsequently, counts within each category were summed and plotted as a horizontal bar chart in ggplot2, with the "Stellate cells" bar highlighted in dark orange and all others shown in light gray. All data processing was done in R (v4.x) using msigdbr, dplyr, and ggplot2.

Real-time quantitative PCR

Liver and adipose tissues were homogenized using TissueLyzer (QIAGEN). Total RNA was extracted using FastPure Cell/Tissue Total RNA Isolation Kit V2 (Cat: RC112-01, Vazyme Biotech Co., Ltd., China) following the manufacturer's protocol. The total RNA concentration was determined using a NanoDrop 2000C spectrophotometer (Thermo Fisher Scientific, Waltham, MA, USA). For each sample, 500 ng of purified RNA was reverse transcribed using the PrimerScript RT Reagent Kit with gDNA eraser (RR047Q, Takara, Kusatsu, Japan). The qPCR primers (Table S4) were designed and synthesized by Tsingke Biotech Co., Ltd. (Shanghai, China).

Quantitative real-time PCR was performed by a Taq Pro Universal SYBR qPCR Master Mix (Cat: Q712-02, Vazyme Biotech Co., Ltd., China) on an ABI Q7 PCR System (Applied Biosystems Instruments, Thermo Fisher Scientific, USA). Target gene expression levels were normalized to *Gapdh*, and relative fold changes were calculated compared to the control group.

Western blot analysis

Liver, intestine, and cell samples were lysed in RIPA buffer (Beyotime Technology, Shanghai, China) supplemented with 1 mM phenylmethylsulfonyl fluoride (PMSF) (Beyotime Technology, Shanghai, China) on ice, followed by centrifugation at 14,000 × *g* for 5 min. The supernatants were collected, and total protein concentration was determined using a BCA Protein Assay Kit (Thermo, California, USA). Protein extracts (5 μg/μl) were mixed with the loading buffer (Beyotime Technology, Shanghai, China), denatured at 100 °C for 10 min, and separated by 10% SDS-page. The resolved proteins were then transferred to a PVDF membrane, which was subsequently blocked with 5% non-fat milk and probed with primary antibodies against target proteins.

Cell culture

The human stellate cell line LX-2 was purchased from FuHeng Cell Center (Shanghai, China). The LX-2 cells were cultured in DMEM high glucose medium (Thermo Fisher Scientific, Waltham, MA, USA) supplemented with 2% fetal bovine serum (FBS, Gibco) and Pen Strep (Gibco, 15140122), incubated at 37 °C containing 5% CO_2 in the air. As for the experiments, LX-2 cells were treated with acetaldehyde (20 μM) alone or with MMP7 inhibitor (MMP-7-IN-2, TargetMol, USA, 0.1 μM) for 24h. Cell samples were harvested for further qPCR and western blot analysis.

Microbial isolation from human feces

Fecal samples were processed for microbiota isolation using four different types of culture media to selectively isolate and cultivate distinct bacterial strains. Initially, approximately 1 g of fecal matter was diluted in 10 mL sterile saline solution and vortexed for 5 minutes to homogenize the sample. The homogenized mixture was then serially diluted to various concentrations (10^{-1} to 10^{-6}) to ensure the appropriate isolation of individual colonies. The diluted samples were then plated onto four distinct selective culture media, each designed to favor the growth of specific microbial populations. The media included: MacConkey agar (for gram-negative, lactose-fermenting bacteria), Brain Heart Infusion agar (BHI agar, for general bacterial growth), De Man, Rogosa and Sharpe agar (MRS agar, for lactic acid bacteria), and Eosin Methylene Blue agar (EMB agar, for enteric bacteria). Plates were incubated at 37 °C for 24 to 48 hours, depending on the targeted microorganisms. After incubation, single bacterial colonies were carefully picked using a sterile loop and transferred to liquid media for further culturing. After obtaining individual colonies, pure cultures were established by subculturing isolated colonies onto fresh agar plates. To ensure monoclonality, streaking methods were repeated until only one colony type was observed. Each pure strain was then identified using 16S rRNA gene sequencing.

Bacterial identification by 16S rRNA gene sequencing

The bacterial strain was cultured on the corresponding medium at 37 °C, and genomic DNA was extracted using TIANamp Bacteria DNA Kit (TIANGEN BIOTECH, Beijing, China). The 16S rRNA gene was amplified via PCR with universal primers (27F/1492R), purified, and sequenced bidirectionally via Sanger sequencing. Sequences were blasted against the NCBI database for taxonomic identification (Table S5).

In vitro assay of the consumption rate of acetaldehyde

The bacterial strain was inoculated into MRS liquid medium and grown overnight at 37 °C. Cells were harvested by centrifugation (4,000 × *g*, 10 min), washed twice with 1× PBS, and then resuspended in fresh 1× PBS. The enzymatic reaction system was prepared by mixing the bacterial suspension with 17 mM ethanol + acetaldehyde or 1 mM acetaldehyde in a 400 μl reaction mixture, followed by incubation at 37 °C for 1 hour. Then the supernatant was collected by centrifugation (16,000 × *g*, 20min). The acetaldehyde levels in the supernatant were assayed using EnzyChrom Acetaldehyde Assay Kit (EACT-100) (BioAssay Systems). A 100 μl aliquot of bacterial suspension was serially diluted in 1× PBS to 10^{-7}-10^{-8}. Each dilution (100 μl) was spread onto MRS agar plates and incubated at 37 °C for 48 hours. Colony-forming units (CFU/mL) were calculated from plate counts (30-300 colonies/plate). The consumption rate of acetaldehyde was then calculated using the determined acetaldehyde level in the supernatant and bacterial concentration in the reaction mixture.

Construction of engineered strain

The genome DNA of *L. salivarius* was extracted using by Rapid Bacterial Genomic DNA Isolation Kit (Sangon Biotech, Shanghai, China) and lysozyme. The aldehyde-alcohol dehydrogenase coding gene was amplified by PCR using the genomic DNA of *L. salivarius* as template. The amplified gene was ligated with pMG36e (an expression vector of *L. salivarius*) that contains elements necessary for the expression of the gene. The ligation mixture was transformed into *Escherichia coli* (*E.coli*) *DH5α* competent cells. The transformed cells were then cultured on Luria-Bertani (LB) plate containing 200 μg/mL erythromycin. The extracted plasmid DNA from single *E.coli* strain was sequenced to identify the correct clone. Subsequently, the amplified recombinant plasmid was electroporated into *L. salivarius*. Transformants were selected on MRS agar plates supplemented with 2.5 μg/mL erythromycin, and positive clones were preserved in glycerol stocks at -80 °C.

Construction of adeno-associated virus (AAV)

As for *Mmp7*-knockdown, AAV8-sh*Control* (pAAV-GfaABC1D-EGFP-3xFLAG-miR30shRNA (NC)-WPRE), AAV8-sh*Mmp7* (AAV-GfaABC1D-EGFP-3xFLAG-miR30shRNA (*Mmp7*)-WPRE) were constructed by Obio Technology, as well as AAV8-sh*Control* (pAAV-TBG-ZsGreen1-miR30shRNA (NC)-WPRE) and AAV8-sh*Aldh2* (pAAV-TBG-ZsGreen1-miR30shRNA (*Aldh2*)-WPRE) for *Aldh2*-knockdown. For AAV virus infection, 2×10^{11} genomic particles were reconstituted in 200 μL PBS and injected intravenously through the tail vein injection with BD ultrafine insulin syringes.

QUANTIFICATION AND STATISTICAL ANALYSIS

Results were presented as mean ± SEM. All bar graphs were generated using GraphPad Prism 10.0 (GraphPad Software, La Jolla, CA, USA). Normality testing was performed using the Kolmogorov-Smirnov test. For normality distributed variables, one-way ANOVA was applied for statistical comparisons. Non-normal distributed data were analyzed using the Kruskal-Wallis test, followed by the Benjamini-Hochberg false discovery rate test performed by SPSS 26.0 (IBM SPSS, Chicago, IL, USA). Correlation analyses between BAs and metabolic parameters were performed using Spearman's correlation analysis, visualized via R studio (RStudio, Boston, MA, USA).